

CHAPTER

37

External and middle ear

By convention, the ear is subdivided into three parts: the external, middle and inner ear. It is largely, but not entirely, contained within the temporal bone. The ears not only receive, modulate, conduct, amplify and discriminately analyse the complex pressure waves that are sound, but also contain the end organs of balance.

TEMPORAL BONE

Each temporal bone consists of four components: the squamous, petromastoid and tympanic parts and the styloid process (Fig. 37.1). The squamous part has a shallow mandibular fossa associated with the temporomandibular joint (Ch. 32). The petromastoid part is relatively large; its petrous portion houses the auditory apparatus and is formed of compact bone. In contrast, the mastoid process is trabecular and variably pneumatized. The tympanic part has the form of a thin and incomplete ring whose ends are fused with the squamous part. The styloid process gives attachment to the styloid group of muscles. The temporal bone contains two canals. The external acoustic meatus (external auditory canal), visible on the lateral surface, conveys sound waves to the tympanic membrane and the internal acoustic meatus (internal auditory canal), evident on the medial surface, through which the facial and vestibulocochlear nerves pass.

Squamous part

The squamous part lies anterosuperiorly, and is thin and partly translucent. Its external temporal surface is smooth, slightly convex, and forms part of the temporal fossa, to which temporalis muscle is attached. Above the external acoustic meatus, it is grooved vertically by the middle temporal artery. The supramastoid crest curves backwards and upwards across its posterior part and gives attachment to the temporal fascia. The junction between the squamous and mastoid parts is approximately 1.5 cm below this crest, and traces of the squamomastoid suture may persist. The suprameatal triangle, a depression marking the position of the mastoid antrum (which is medial to the triangle at a depth of approximately 1.25 cm), lies between the anterior end of the supramastoid crest and the posterosuperior quadrant of the external acoustic meatus. The triangle usually contains a small suprameatal spine anteriorly.

The internal cerebral surface of the squamous part is concave and contains depressions that correspond to convolutions of the temporal lobe of the cerebral hemisphere. This surface is grooved by the middle meningeal vessels. Its lower border is fused to the anterior region of the petrous part but traces of a petrosquamosal suture often appear in adult bones. The superior border is thin and bevelled internally; it overlaps the inferior border of the parietal bone at the squamosal suture. Posteriorly, it forms an angle with the mastoid element. The anteroinferior border, thin above and thick below, meets the greater wing of the sphenoid bone; it is bevelled internally above, and bevelled externally below.

The squamous part has a zygomatic process and a mandibular fossa.

Zygomatic process

The zygomatic process juts forwards from the lower region of the squamous part. Its triangular posterior aspect has a broad base that is directed laterally, presenting superior and inferior surfaces. The zygomatic process then twists anteromedially, so that its surfaces become medial and lateral.

The superior surface of the posterior part is concave. The inferior surface is bounded by anterior and posterior roots, converging into the anterior part of the process. The tubercle of the zygomatic root gives attachment to the lateral temporomandibular ligament at the junction of the roots. The posterior root is prolonged forwards above the external acoustic meatus, its upper border continuing into the supramastoid crest. Very rarely, the squamous part is perforated above the posterior

root by a squamosal foramen, which transmits the petrosquamous sinus. The anterior root juts almost horizontally from the squamous part. Its inferior surface, with an anteroposterior convexity, forms a short semi-cylindrical articular tubercle and comes into contact with the articular disc of the temporomandibular joint. The tubercle forms the anterior limit of the mandibular fossa.

The anterior part of the zygomatic process is thin and flat, and the temporal fascia is attached to its superior border. The inferior border is short and arched, and gives origin to some fibres of masseter. The lateral surface is convex. The medial surface is concave and provides further attachment for part of masseter. The anterior end is deeply serrated and slopes obliquely posteroinferiorly to articulate with the temporal process of the zygomatic bone, forming the zygomatic arch. Anterior to the articular tubercle, a small triangular area forms part of the roof of the infratemporal fossa; it is continuous behind with the anterior root and in front with the infratemporal crest of the greater wing of the sphenoid.

Mandibular fossa

The mandibular fossa is limited in front by the articular eminence of the zygomatic process. It presents an anterior articular area, formed by the squamous part, and a posterior non-articular area, formed by the tympanic element. The articular surface is smooth, oval and concave, and contacts the articular disc of the temporomandibular joint. Unlike most other synovial joints, it is lined by fibrous tissue rather than hyaline cartilage, reflecting its intramembranous development. The non-articular area sometimes contains part of the parotid gland. A small, conical postglenoid tubercle separates the articular surface laterally from the tympanic plate.

Posteriorly, the mandibular fossa is separated from the tympanic part by the squamotympanic fissure. Rarely, a postglenoid foramen exists anterior to the external acoustic meatus in the line of fusion of the squamous and tympanic parts. When present, it replaces the squamosal foramen noted above and transmits the petrosquamous sinus. Medially, a projection from the petrous part of the temporal bone (tegmen tympani) comes to lie within the squamotympanic fissure, further dividing it into petrotympanic and petrosquamous fissures. The petrotympanic fissure leads into the tympanic cavity and contains an anterior malleolar ligament and the anterior tympanic branch of the maxillary artery. The anterior opening of the anterior canaliculus for the chorda tympani nerve lies at the medial end of the fissure.

Petromastoid part

The petromastoid part of the temporal bone, although morphologically one element, is more conveniently described as two parts, namely: mastoid and petrous parts.

Mastoid part

The mastoid part is the posterior region of the temporal bone and has an outer surface roughened by the attachments of the occipital belly of occipitofrontalis and auricularis posterior. A mastoid foramen, of variable size and position, and traversed by a vein from the sigmoid sinus and a small dural branch of the occipital artery, frequently lies near its posterior border. The foramen may be in the occipital or occipitotemporal suture; it may sometimes be parasutural or may be absent.

The mastoid part projects down as the conical mastoid process and is larger in adult males. Sternocleidomastoid, splenius capitis and longissimus capitis are all attached to its lateral surface, and the posterior belly of digastric is attached to a deep mastoid notch on its medial aspect. The occipital artery runs in a shallow occipital groove that lies medial to the mastoid notch. The internal surface of the mastoid process bears a deep, curved sigmoid sulcus for the sigmoid venous sinus; the sulcus is separated from the underlying innermost mastoid air cells by a thin lamina of bone.

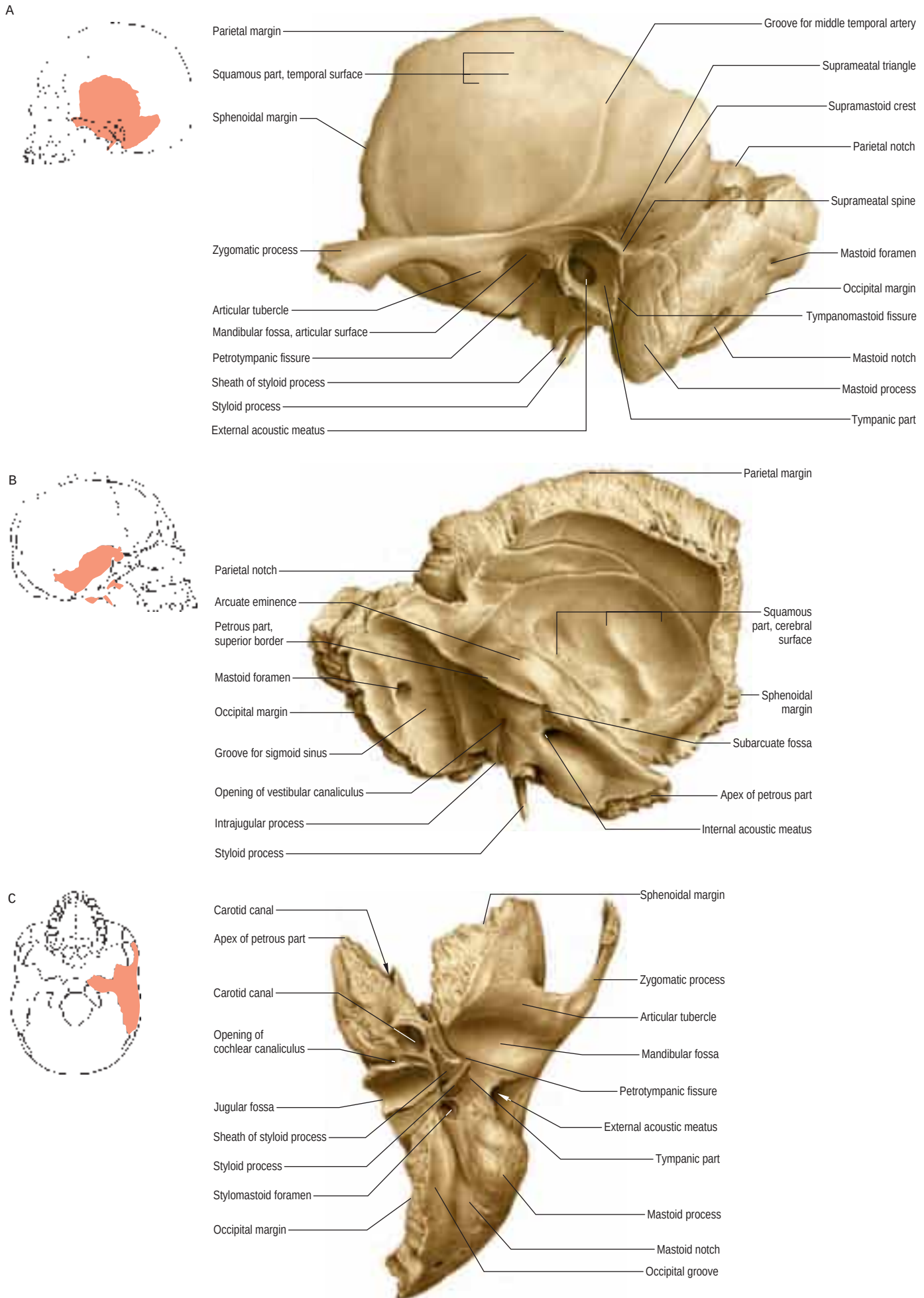


Fig. 37.1 The left temporal bone. **A**, Lateral aspect. **B**, Medial aspect. **C**, Inferior aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

The superior border of the mastoid part is thick and serrated for articulation with the mastoid angle of the parietal bone. The posterior border is also serrated and articulates with the inferior border of the occipital bone between its lateral angle and jugular process. The mastoid element is fused with the descending process of the squamous part; below, it appears in the posterior wall of the tympanic cavity.

Petrous part

The petrous part is a mass of bone that is wedged between the sphenoid and occipital bones in the cranial base; it contains the labyrinth. It is inclined superiorly and anteromedially, and has a base, apex, three surfaces (anterior, posterior and inferior) and three borders (superior, posterior and anterior).

The base would correspond to the part that lies on the base of the skull and is separated from the squamous part by a suture. However, this suture disappears soon after birth. The subsequent development of the mastoid processes means that the precise boundaries of the base are no longer identifiable.

The apex, blunt and irregular, is angled between the posterior border of the greater wing of the sphenoid and the basilar part of the occipital bone. It contains the anterior opening of the carotid canal and limits the foramen lacerum posterolaterally.

The anterior surface contributes to the floor of the middle cranial fossa (Ch. 28) and is continuous with the cerebral surface of the squamous part (although the petrosquamosal suture often persists late in life). The whole surface is adapted to the inferior temporal gyrus. Behind the apex is a trigeminal impression for the trigeminal ganglion. Bone anterolateral to this impression roofs the anterior part of the carotid canal but is often deficient. A ridge separates the trigeminal impression from another hollow behind, which partly roofs the internal acoustic meatus and cochlea. This, in turn, is limited behind by the arcuate eminence, which is raised by the superior (anterior) semicircular canal but is not necessarily directly over it. Laterally, the anterior surface roofs the vestibule and, partly, the facial canal. Between the squamous part laterally and the arcuate eminence and the hollows just described medially, the anterior surface is formed by the tegmen tympani, a thin plate of bone that forms the roof of the mastoid antrum, and extends forwards above the tympanic cavity and the canal for tensor tympani. The lateral margin of the tegmen tympani meets the squamous part at the petrosquamosal suture, turning down in front as the lateral wall of the canal for tensor tympani and the osseous part of the pharyngotympanic tube; its lower edge is in the squamotympanic fissure. Anteriorly, the tegmen bears a narrow groove related to the greater petrosal nerve (which passes posterolaterally to enter the bone by a hiatus anterior to the arcuate eminence). The groove passes forwards to the foramen lacerum. A smaller and similar hiatus and groove may be found more laterally; they are related to the lesser petrosal nerve (which runs to the foramen ovale). The posterior slope of the arcuate eminence overlies the posterior and lateral semicircular canals. Lateral to the eminence, the posterior part of the tegmen tympani roofs the mastoid antrum.

The posterior surface contributes to the anterior part of the posterior cranial fossa and is continuous with the internal surface of the mastoid part. The opening of the internal acoustic meatus lies near its centre. A small slit leading to the vestibular aqueduct lies behind the opening of the meatus, almost hidden by a thin plate of bone. This contains the saccus and ductus endolymphaticus, together with a small artery and vein. The terminal half of the saccus endolymphaticus protrudes through the slit between the periosteum and dura mater. The subarcuate fossa lies above these openings.

The irregular inferior surface is part of the exterior of the cranial base. Near the apex of the petrous part, a quadrilateral area is partly associated with the attachment of levator veli palatini and the cartilaginous pharyngotympanic tube, and partly connected to the basilar part of the occipital bone by dense fibrocartilage. Behind this region is the large, circular opening of the carotid canal, and behind the opening of the canal is the jugular fossa, which is of variable depth and size, and contains the superior jugular bulb. The inferior ganglion of the glossopharyngeal nerve lies in a triangular depression anteromedial to the jugular fossa (below the internal acoustic meatus). At its apex is a small opening into the cochlear canaliculus, occupied by the perilymphatic duct (a tube of dura mater) and a vein draining from the cochlea to the internal jugular vein. A canaliculus for the tympanic nerve from the glossopharyngeal nerve lies on the ridge between the carotid canal and the jugular fossa. The mastoid canaliculus for the auricular branch of the vagus nerve is laterally positioned in the jugular fossa. Behind the jugular fossa, the rough quadrilateral jugular surface is covered by cartilage that joins it to the jugular process of the occipital bone.

The superior border, the longest, is grooved by the superior petrosal sinus. The attached margin of the tentorium cerebelli is fixed to the edges of the groove except at its medial end, where it is crossed by the roots of the trigeminal nerve. The posterior border, intermediate in length, bears a sulcus medially, which forms, together with the occipital bone, a gutter for the inferior petrosal sinus. Behind this, the jugular fossa contributes (together with the occipital bone) to the jugular foramen and is notched by the glossopharyngeal nerve. Bone on either or both sides of the jugular notch may meet the occipital bone and divide the jugular foramen into two or three parts. The anterior border is joined laterally to the squamous part of the temporal bone at the petrosquamosal suture; medially, it articulates with the greater wing of the sphenoid bone.

Two canals exist at the junction of the petrous and squamous parts, one above the other, separated by a thin osseous plate and both leading to the tympanic cavity; the upper canal contains tensor tympani, while the lower canal is the pharyngotympanic tube.

Tympanic part

The tympanic part of the temporal bone is a curved plate below the squamous part and anterior to the mastoid process. Internally, it fuses with the petrous part and appears between this and the squamous part, where it is inferolateral to the auditory orifice. Behind, it fuses with the squamous part and mastoid process, and is the anterior limit of the tympanomastoid fissure. Its concave posterior surface forms the anterior wall, floor and part of the posterior wall of the external acoustic meatus. The tympanic membrane is attached to a narrow tympanic sulcus on its medial surface. The quadrilateral concave anterior surface is the posterior wall of the mandibular fossa and may contact the parotid gland. Its rough lateral border forms most of the margin of the osseous part of the external acoustic meatus and is continuous with its cartilaginous part. Laterally, the upper border is fused with the back of the postglenoid tubercle; medially, it forms the posterior edge of the petrotympanic fissure. The inferior border is sharp and splits laterally to form, at its root, the sheath of the styloid process (vaginal process). Centrally, the tympanic part is thin and is often perforated. The stylo-mastoid foramen lies between the styloid and mastoid processes; it represents the external end of the facial canal, contains part of the aponeurosis of the posterior belly of digastric, and transmits the facial nerve and stylomastoid artery.

Styloid process

The styloid process is slender and pointed, and projects anteroinferiorly from the inferior aspect of the temporal bone. Its length varies, ranging from a few millimetres to an average of 2.5 cm. Often almost straight, it can show a curvature, an anteromedial concavity being most common. Its proximal part (tyimpanohyal) is ensheathed by the tympanic plate, especially anterolaterally, while muscles and ligaments are attached to its distal part (stylohyal). *In vivo*, its relationships are important. The styloid process is covered laterally by the parotid gland; the facial nerve crosses its base; the external carotid artery crosses its tip, embedded in the parotid; and medially, the process is separated from the beginning of the internal jugular vein by the attachment of stylopharyngeus.

External acoustic meatus

The temporal bone contains the bony (osseous) part of the external acoustic meatus.

Ossification

The four temporal components ossify independently (Fig. 37.2). The squamous part is ossified in a sheet of condensed mesenchyme from a single centre near the zygomatic roots, which appears in the seventh or eighth week *in utero*. The petromastoid part has several centres that appear in the cartilaginous otic capsule during the fifth month; as many as 14 have been described. These centres vary in order of appearance. Several are small and inconstant, soon fusing with others. The otic capsule is almost fully ossified by the end of the sixth month. The tympanic part is also ossified in mesenchyme from a centre identifiable about the third month; at birth, it is an incomplete tympanic ring, deficient above, its concavity grooved by a tympanic sulcus for the tympanic membrane. The malleolar sulcus for the anterior malleolar process, chorda tympani and anterior tympanic artery inclines obliquely downwards and forwards across the medial aspect of the anterior part

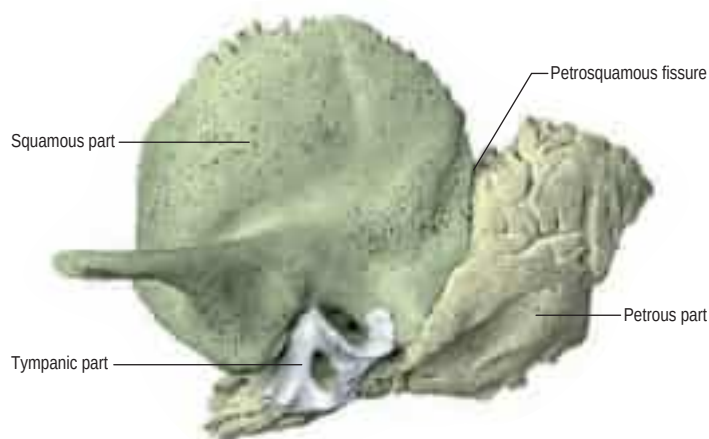


Fig. 37.2 The left temporal bone at birth. (With permission from Waschke J, Paulsen F (eds), *Sobotta Atlas of Human Anatomy*, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

of the ring. The styloid process develops from two centres at the cranial end of cartilage in the second visceral or hyoid arch; a proximal centre for the tympanohyal appears before birth, and another, for the distal stylohyal, appears after birth. The tympanic ring unites with the squamous part shortly before birth, and the petromastoid fuses with it and the tympanohyal during the first year. The stylohyal does not unite with the rest of the process until after puberty and may never do so.

Once ossified, the tympanic cavity, mastoid antrum and the posterior end of the pharyngotympanic tube become surrounded by bone. The petrous part forms the roof, floor and medial wall of the cavity, while the squamous and tympanic parts, together with the tympanic membrane, form its lateral wall. At birth, the middle and inner ears are adult size, and the tympanic cavity, mastoid antrum, tympanic membrane and auditory ossicles are all almost adult size. The anterior process does not join the malleus until 6 months later. The internal acoustic meatus is approximately 6 mm in horizontal diameter, 4 mm in vertical diameter and 7 mm in length at birth, and the adult diameters are 7.7 mm and 11 mm, respectively.

After birth and apart from general growth, the tympanic ring extends posterolaterally to become cylindrical, growing into a fibrocartilaginous tympanic plate, which forms the adjacent part of the external acoustic meatus at this stage. This growth is not equal but is rapid in the anterior and posterior regions, which meet and blend. Thus, for a time, an opening (foramen of Huschke) exists in the floor; it usually closes at about the fifth year but is sometimes permanent. The external acoustic meatus is relatively as long in children as it is in adults, but the canal is fibrocartilaginous, whereas its medial two-thirds are osseous in adults. Surgical access to the tympanic cavity is via the mastoid antrum, and in children it is necessary to remove only a thin scale of bone in the suprameatal triangle to reach the antrum. The tympanic plate ensheathes the styloid process by posterior extension, and extends medially over the petrous bone to the carotid canal.

Initially, the mandibular fossa is shallow, facing more laterally, but it then deepens and ultimately faces downwards. Posteroinferiorly, the squamous part grows down behind the tympanic ring to form the lateral wall of the mastoid antrum. The mastoid part is at first flat, so that the stylo-mastoid foramen and rudimentary styloid process are immediately behind the tympanic ring. The mastoid part becomes invaded by air cells, especially at puberty. The lateral mastoid region grows downwards and forwards to form the mastoid process, which means that the styloid process and stylo-mastoid foramen become inferior. Descent of the foramen lengthens the facial canal. The mastoid process is not perceptible until late in the second year; consequently, the facial nerve is relatively superficial and susceptible to damage during surgical exploration during this period. The subarcuate fossa gradually fills and is almost obliterated.

In the neonate, the petrous and squamous parts of the temporal bone are usually partially separated by the petrosquamous fissure, which opens directly into the mastoid antrum of the middle ear. Rarely, the fissure closes in infants during the first year but it sometimes remains unclosed up to the age of 19 years; it is a route for the spread of infection from the middle ear to the meninges. There is a bimodal growth pattern in the lateral surface dimensions of the temporal bone in children; marked increases in dimension occur from birth to the age of 4 years but little growth is seen between the ages of 4 and 20 years (Simms and Neely 1989). The neonatal internal acoustic meatus is about half the length of its adult counterpart. Its opening from the



Fig. 37.3 The lateral surface of the left auricle. Key: 1, helix; 2, crus of helix; 3, auricular tubercle; 4, antihelix; 5, crura of antihelix; 6, triangular fossa; 7, scaphoid fossa; 8, concha of auricle; 9, external acoustic meatus; 10, tragus; 11, antitragus; 12, intertragic notch; 13, lobule of auricle. (With permission from Berkovitz BKB, Moxham BJ 2002 *Head and Neck Anatomy*. London: Martin Dunitz.)

middle ear cavity is as large as it is in the adult but the pharyngeal opening in the nasal part of the pharynx is relatively smaller. The course of the pharyngotympanic tube is horizontal in the newborn, whereas in the adult it passes from the middle ear downwards, forwards and medially.

EXTERNAL EAR

The external ear consists of the auricle, or pinna, and the external acoustic meatus. The auricle projects to a variable and individual degree from the side of the head. It collects sound waves and conducts them along the external acoustic meatus inwards towards the ear drum, the tympanic membrane. The auricle thus acts as the first element of a series of stimulus modifiers in the auditory apparatus.

AURICLE (PINNA)

The lateral surface of the auricle is irregularly concave, faces slightly forwards, and displays numerous eminences and depressions (Fig. 37.3). It has a prominent curved rim, the helix. This usually bears a small tubercle posterosuperiorly, Darwin's tubercle, which is quite pronounced around the sixth month of intrauterine life. The antihelix is a curved prominence, parallel and anterior to the posterior part of the helix; it divides above into two crura, which flank a depressed triangular fossa. The curved depression between the helix and antihelix is the scaphoid fossa. The antihelix encircles the deep, capacious concha of the auricle, which is incompletely divided by the crus or anterior end of the helix. The conchal area above this, the cymba conchae, overlies the suprameatal triangle of the temporal bone, which can be felt through it, and which overlies the mastoid antrum. The tragus is a small curved flap below the crus of the helix and in front of the concha; it projects posteriorly, partly overlapping the meatal orifice. The antitragus is a small tubercle opposite the tragus and is separated from it by the intertragic incisure or notch. Below it is the lobule, composed of fibrous and adipose tissues. It is soft, unlike the majority of the auricle, which is supported by elastic cartilage and is firm. The cranial surface of the auricle presents elevations that correspond to the depressions on its lateral surface and after which they are named (e.g. eminentia conchae, eminentia fossae triangularis).

At birth, most of the linear dimensions of the auricle are approximately three-quarters of their adult size; the length and height of the tragus are less than half of their adult size. Width dimensions mature between the ages of 5 and 11 years; length dimensions mature between 12 and 16 years (Purkait 2013).

Common congenital anomalies

Developmental anomalies of the branchial arches may produce a grossly misshapen or microtic auricle, sometimes with associated anomalies of the middle ear and significant hearing loss. Deformity of the pinna may be the result of hereditary factors, exposure to teratogens



or unusual intrauterine positioning. A number of common anomalies have been recognized; they carry descriptive names or eponyms (Porter and Tan 2005) (Table 37.1).

Skin

The skin of the auricle continues into the external acoustic meatus to cover the outer surface of the tympanic membrane. It is thin, has no dermal papillae, and is closely adherent to the cartilaginous and osseous parts of the canal; inflammation of the canal skin is very painful because of this attachment to the underlying structures. The thick subcutaneous tissue of the cartilaginous part of the meatus contains numerous ceruminous glands that secrete wax, or cerumen. Their coiled tubular structure resembles that of sweat glands. The secretory cells are columnar when active but cuboidal when quiescent; they are covered externally by myoepithelial cells. Ducts open either on to the epithelial surface or into the nearby sebaceous gland of a hair follicle. Cerumen prevents the maceration of meatal skin by trapped water. Antibacterial properties have been attributed to cerumen but the evidence for this is lacking (Campos et al 2000, Pata et al 2003).

Two types of wax – wet and dry – are recognized. They are genetically determined. Dry wax is common in East Asians, while the wet type is more common in other ethnic groups (Yoshiura et al 2006). Overproduction, accumulation or impaction of wax may completely occlude the meatus. This may hinder sound from reaching the tympanic membrane and so restrict its natural vibration. Although ceruminous glands and hair follicles are largely limited to the cartilaginous meatus, a few small glands and fine hairs are also present in the roof of the lateral part of the bony meatus. The warm, humid environment of the relatively enclosed meatal air aids the mechanical responses of the tympanic membrane.

Cartilaginous framework

The auricle is a single thin plate of elastic fibrocartilage covered by skin, its surface moulded by eminences and depressions (Fig. 37.4). It is connected to the surrounding parts by ligaments and muscles, and is continuous with the cartilage of the external acoustic meatus.

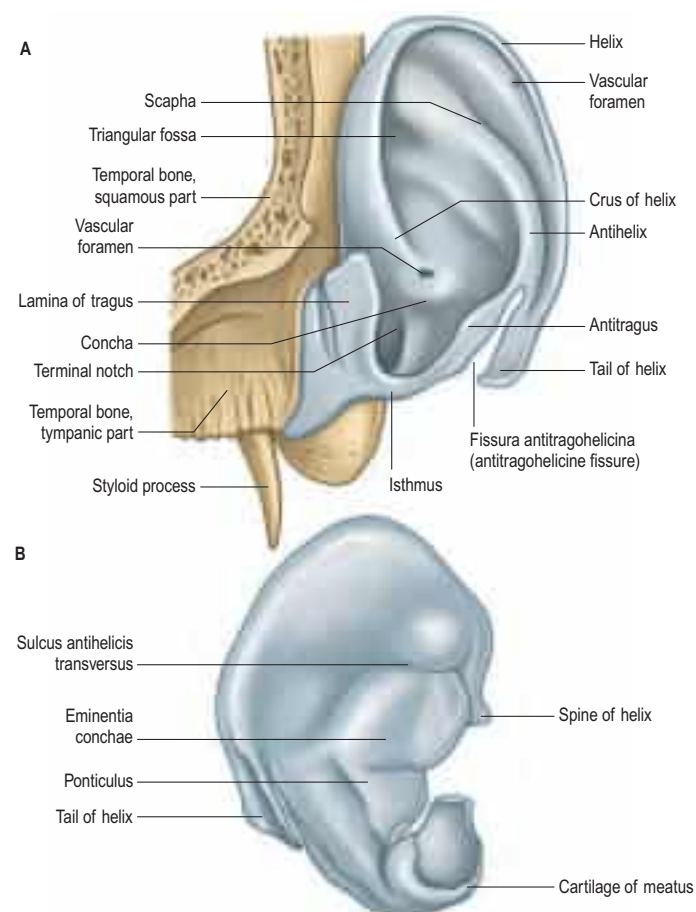


Fig. 37.4 The auricular cartilages of the left ear. **A**, Lateral surface. **B**, Medial surface.

is no cartilage in the lobule or between the tragus and the crus of the helix, where the gap is filled by dense fibrous tissue. Anteriorly, where the helix curves upwards, there is a small cartilaginous projection: the spine of the helix. Its other extremity is prolonged inferiorly as the tail of the helix and it is separated from the antihelix by the fissura antitragohelicina (antitragohelicine fissure). The cranial aspect of the cartilage bears the eminentia conchae and eminentia scaphae, which correspond to the depressions on the lateral surface. The two eminences are separated by a transverse furrow, the sulcus antihellicus transversus, which corresponds to the inferior crus of the antihelix on the lateral surface. The eminentia conchae is crossed by an oblique ridge, the ponticulus, for the attachment of auricularis posterior. There are two fissures in the auricular cartilage, one behind the crus of the helix and another in the tragus.

Ligaments

Anterior and posterior extrinsic ligaments connect the auricle with the temporal bone. The anterior ligament extends from the tragus and the spine of the helix to the root of the zygomatic process of the temporal bone. The posterior ligament passes from the posterior surface of the concha to the lateral surface of the mastoid process. Two main intrinsic ligaments connect individual auricular cartilages: a strong fibrous band passes from the tragus to the helix, thereby completing the meatus anteriorly and forming part of the boundary of the concha; and another band passes between the antihelix and the tail of the helix. Less prominent bands exist on the cranial aspect of the auricle.

Auricular muscles

Extrinsic auricular muscles connect the auricle to the skull and scalp, and move the auricle as a whole. Intrinsic auricular muscles connect the different parts of the auricle.

Extrinsic muscles

The extrinsic auricular muscles are the auriculares anterior, superior and posterior (Fig. 37.5B). The smallest of the three is auricularis anterior, a thin fan of pale fibres that arise from the lateral edge of the epicranial aponeurosis and converge to attach to the spine of the helix. The largest of the three, auricularis superior, is also thin and fan-shaped, and converges from the epicranial aponeurosis via a thin, flat tendon to attach to the upper part of the cranial surface of the auricle. The auricularis posterior consists of two or three fleshy fasciculi that arise by short aponeurotic fibres from the mastoid part of the temporal bone and insert into the ponticulus on the eminentia conchae.

Vascular supply The arterial supply of the extrinsic auricular muscles is derived mainly from the posterior auricular artery.

Innervation Auriculares anterior and superior are supplied by temporal branches of the facial nerve, and auricularis posterior is supplied by the posterior auricular branch of the facial nerve.

Actions In humans, these muscles have very little obvious effect. Despite the paucity of auricular movement, auditory stimuli may evoke patterned responses from these small muscles and electromyography can detect the 'crossed acoustic response' that can be used to determine auditory threshold levels and brainstem latencies, which is elicited by this means in investigative clinical neurotology.

Intrinsic muscles

The intrinsic auricular muscles are helices major and minor, tragus, antitragicus, transversus auriculæ and obliquus auriculæ (Fig. 37.5A). Helices major is a narrow vertical band on the anterior margin of the helix, passing from its spine to its anterior border, where the helix is about to curve back. Helices minor is an oblique fasciculus covering the crus of the helix. Tragus is a short, flattened, vertical band on the lateral aspect of the tragus. Antitragicus passes from the outer part of the antitragus to the tail of the helix and the antihelix. Transversus auriculæ, located on the cranial aspect of the auricle, consists of scattered fibres – partly tendinous, partly muscular, which extend between the eminentia conchae and the eminentia scaphae. Obliquus auriculæ, also located on the cranial aspect of the auricle, consists of a few fibres that extend from the upper and posterior parts of the eminentia conchae to the eminentia scaphae.

Vascular supply The intrinsic auricular muscles are supplied by branches of the posterior auricular and superficial temporal arteries.

Table 37.1 Common anomalies of the auricle

Anomaly	Structural defect	Figure
Anotia	Complete absence of the external ear that is most likely caused by a developmental disturbance between the seventh and eighth gestational weeks	
Cryptotia (Hidden ear, pocket ear)	The upper pole of the pinna is buried beneath the temporal skin	
Microtia	Diminutive ear; this is usually an isolated congenital anomaly but is sometimes associated with recognized syndromes, e.g. fetal alcohol syndrome, maternal diabetic syndrome, thalidomide or isotretinoin exposure	

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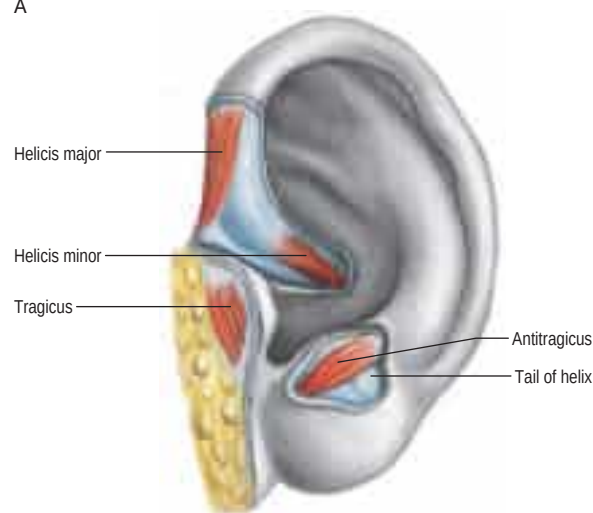
Table 37.1 Common anomalies of the auricle—cont’d

Anomaly	Structural defect	Figure
Polyotia (Mirror ear)	Caused by the persistence of pre-auricular tissue that would normally be included in the pinna; the persistent tissue lies in front of the tragus in the posterior aspect of the cheek	
Stahl's bar ('Satiro's ear', 'lop ear')	Common congenital deformity of the auricle in which the helix is flattened and the upper crus of the antihelix is duplicated. This produces a ridge of cartilage running from the antihelix to the rim of the helix, causing a pointing of the ear and a reversal of the normal concavity of the scaphoid fossa. The upper part of the pinna may flop over	
Prominent ears (Bat ear)	The antihelical fold is either absent or inadequate	

Table 37.1 Common anomalies of the auricle—cont'd

Anomaly	Structural defect	Figure
Pre-auricular sinus	The result of failure of fusion of the first and second branchial arches. Clinically, pre-auricular sinuses may become chronically infected and require surgical excision. This may be technically demanding, given the close proximity to the facial nerve and auricular tubercles around the dorsal end of the first branchial cleft. The sinuses may be simple pits or complex branching sinuses that occasionally extend deeply towards the external acoustic meatus so that they lie close to the facial nerve. There is debate as to whether the anomalies are epithelial inclusions between the hillocks or remnants of the first branchial cleft	

A



B

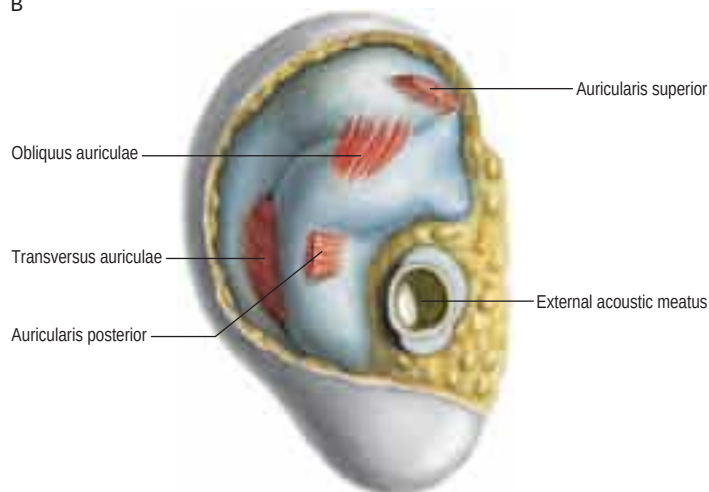


Fig. 37.5 Extrinsic and intrinsic auricular muscles of the left ear. Auricularis anterior is not illustrated. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

Innervation The intrinsic auricular muscles on the lateral aspect of the auricle are innervated by the temporal branches of the facial nerve, and those on the cranial aspect of the auricle are innervated by the posterior auricular branch of the facial nerve.

Actions The intrinsic muscles modify auricular shape minimally, if at all, in most human ears: helicis major can draw the auricle forwards and upwards. Rare individuals can modify the shape and position of their external ears.

Vascular supply and lymphatic drainage

Arteries The posterior auricular branch of the external carotid artery is the dominant blood supply (Imanishi et al 1997). It supplies three or four branches to the cranial surface of the auricle; twigs from these arteries reach the lateral surface – some through fissures in the cartilage, others round the margin of the helix. The posterior auricular artery ascends between the parotid gland and the styloid process to the groove between the auricular cartilage and mastoid process. The superior auricular artery has a constant course and connects the superior temporal artery and the posterior auricular arterial network; this branch can provide a reliable vascular pedicle for retro-auricular flaps (Moschella et al 2003). The auricle is also supplied by anterior auricular branches of the superficial temporal artery, which are distributed to its lateral surface, and by a branch from the occipital artery.

Veins Auricular veins correspond to the arteries of the auricle. Arterio-venous anastomoses are numerous in the skin of the auricle and are thought to be important in the regulation of core temperature.

Lymphatic drainage The posterior aspect of the pinna drains to nodes at the mastoid tip. The tragus and upper part of the pinna drain

into pre-auricular nodes, while the remainder of the pinna drains to upper deep cervical lymph nodes.

Innervation

The sensory innervation of the auricle is complex and not fully determined. This is perhaps because the external ear represents an area where skin originally derived from a branchial region meets skin originally derived from a postbranchial region. The sensory nerves involved are the great auricular nerve, which supplies most of the cranial surface and the posterior part of the lateral surface (helix, antihelix, lobule); the lesser occipital nerve, which supplies the upper part of the cranial surface; the auricular branch of the vagus, which supplies the concavity of the concha and posterior part of the eminentia; the auriculotemporal nerve, which supplies the tragus, crus of the helix and the adjacent part of the helix; and the facial nerve, which, together with the auricular branch of the vagus, probably supplies small areas on both aspects of the auricle, in the depression of the concha and over its eminence. The details of the cutaneous innervation derived from the facial nerve require further clarification. It is possible that, as the auricular branch of the vagus traverses the temporal bone and crosses the facial canal, approximately 4 mm above the stylomastoid foramen, it contributes an ascending branch to the facial nerve and that, in this way, fibres of the vagus are carried via the facial nerve to the pinna.

EXTERNAL ACOUSTIC MEATUS

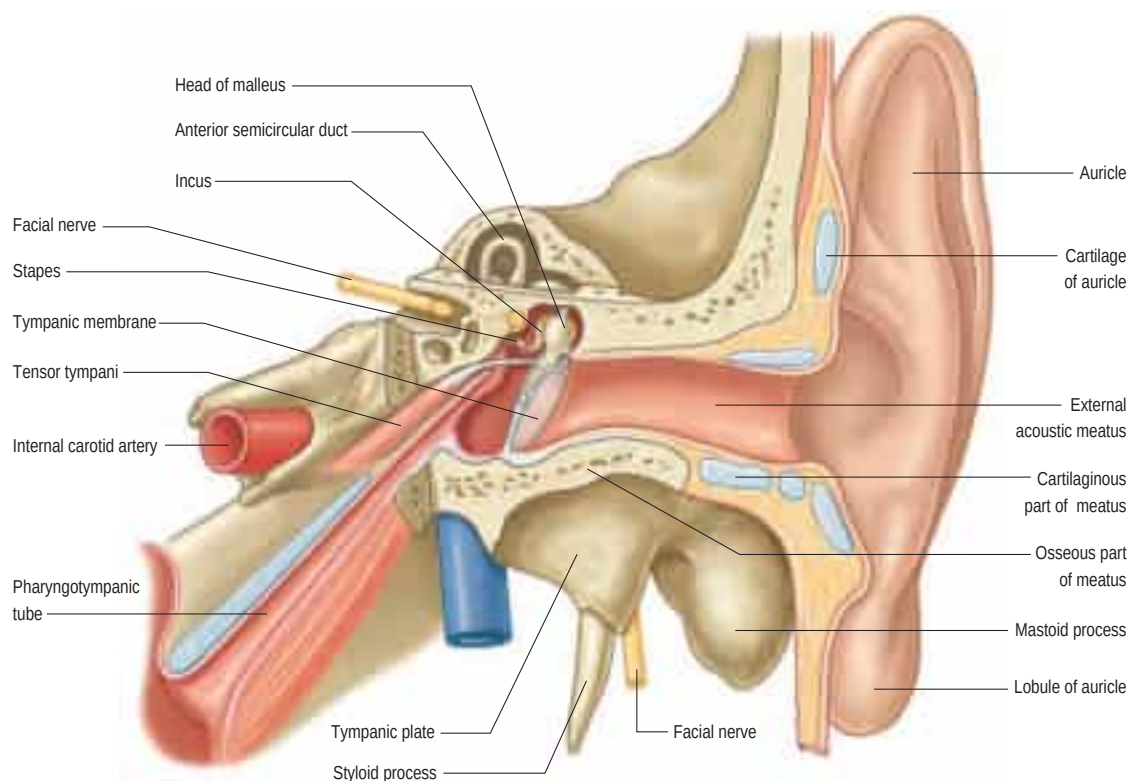
The external acoustic meatus extends from the concha to the tympanic membrane; it is approximately 2.5 cm from the floor of the concha and approximately 4 cm from the tragus. It has two structurally different parts: its lateral third is cartilaginous and its medial two-thirds is osseous (Figs 37.6–37.8). It forms an S-shaped curve, directed at first medially, anteriorly and slightly up (pars externa), then posteromedially and up (pars media), and lastly anteromedially and slightly down (pars interna). It is oval in section; its greatest diameter is obliquely inclined posteroinferiorly at the external orifice but is nearly horizontal at its medial end. There are two constrictions: one near the medial end of the cartilaginous part, and the other, the isthmus, in the osseous part about 2 cm from the bottom of the concha. The tympanic membrane, which closes its medial end, is obliquely set, which means that the floor and the anterior wall of the meatus are longer than its roof and posterior wall.

The lateral, cartilaginous part is approximately 8 mm long. It is continuous with the auricular cartilage and attached by fibrous tissue to the circumference of the osseous part. The meatal cartilage is deficient posterosuperiorly, and the gap is occupied by a sheet of collagen. Two or three deep fissures (of Santorini) exist in its anterior part. Tumours of the external acoustic meatus escape the confines of the canal through these fissures and spread into the adjacent soft tissues; the extent of these fissures, which are subject to individual variation, must influence outcomes.

The osseous part is approximately 16 mm long and is narrower than the cartilaginous part. In sagittal section, it is oval or elliptical, and it is directed anteromedially and slightly downwards, with a slight posterosuperior convexity. Its medial end is smaller than the lateral end and it terminates obliquely. The anterior wall projects medially approximately 4 mm beyond the posterior and is marked, except above, by a narrow tympanic sulcus or anulus, to which the perimeter of the tympanic membrane is attached. Its lateral end is dilated and mostly rough for the attachment of the meatal cartilage. The anterior, inferior, and most of the posterior parts of the osseous meatus are formed by the tympanic plate of the temporal bone, which in the fetus is only a tympanic ring. The posterosuperior region is formed by the squamous part of the temporal bone. The outer wall of the meatus is bounded above by the posterior zygomatic root, below which there may be a suprameatal spine.

Relations of the meatus The condylar process of the mandible lies anterior to the meatus and is partially separated from the cartilaginous part by a small portion of the parotid gland. A blow on the chin may cause the condyle to break into the meatus. The middle cranial fossa lies above the osseous meatus and the mastoid air cells are posterior to it, separated from the meatus only by a thin layer of bone. Its deepest part is situated below the epitympanic recess and is anteroinferior to the mastoid antrum; the lamina of bone that separates it from the antrum is only 1–2 mm thick and provides the ‘transmeatal approach’ of aural surgery.

A



B

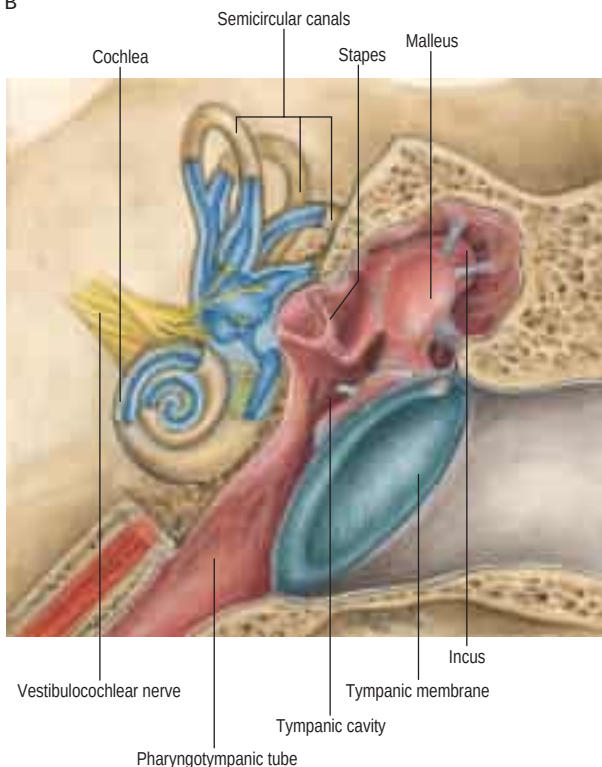


Fig. 37.6 The external, middle and inner regions of the left ear. **A**, Anterior aspect. **B**, A more detailed view of the relationships of structures in the middle and inner ear. (B, With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

Vasculature and lymphatic drainage The arterial supply of the external acoustic meatus is derived from the posterior auricular artery, the deep auricular branch of the maxillary artery and the auricular branches of the superficial temporal artery. Associated veins drain into the external jugular and maxillary veins and the pterygoid plexus. The lymphatics drain into those associated with the pinna.

Innervation The sensory innervation of the external acoustic meatus is derived from the auriculotemporal branch of the mandibular nerve (see Fig. 32.23), which supplies the anterior and superior walls, and the auricular branch of the vagus, which supplies the posterior and inferior walls. The facial nerve may also contribute via its communication with the vagus nerve.

EXTERNAL SURGICAL APPROACHES TO THE MIDDLE EAR

Surgical access to the middle ear can be achieved by a number of methods. Provided the external acoustic meatus is wide enough, the tympanic membrane can be elevated by incising the skin of the bony meatus circumferentially, leaving a vascular pedicle superiorly. The canal skin is then elevated from the underlying bone until the fibrous annulus of the tympanic membrane is visualized. This can then be elevated from the tympanic groove and the middle ear mucosa can be incised to allow the tympanic membrane to be reflected forwards and upwards. This per-meatal approach is called a tympanotomy and is used

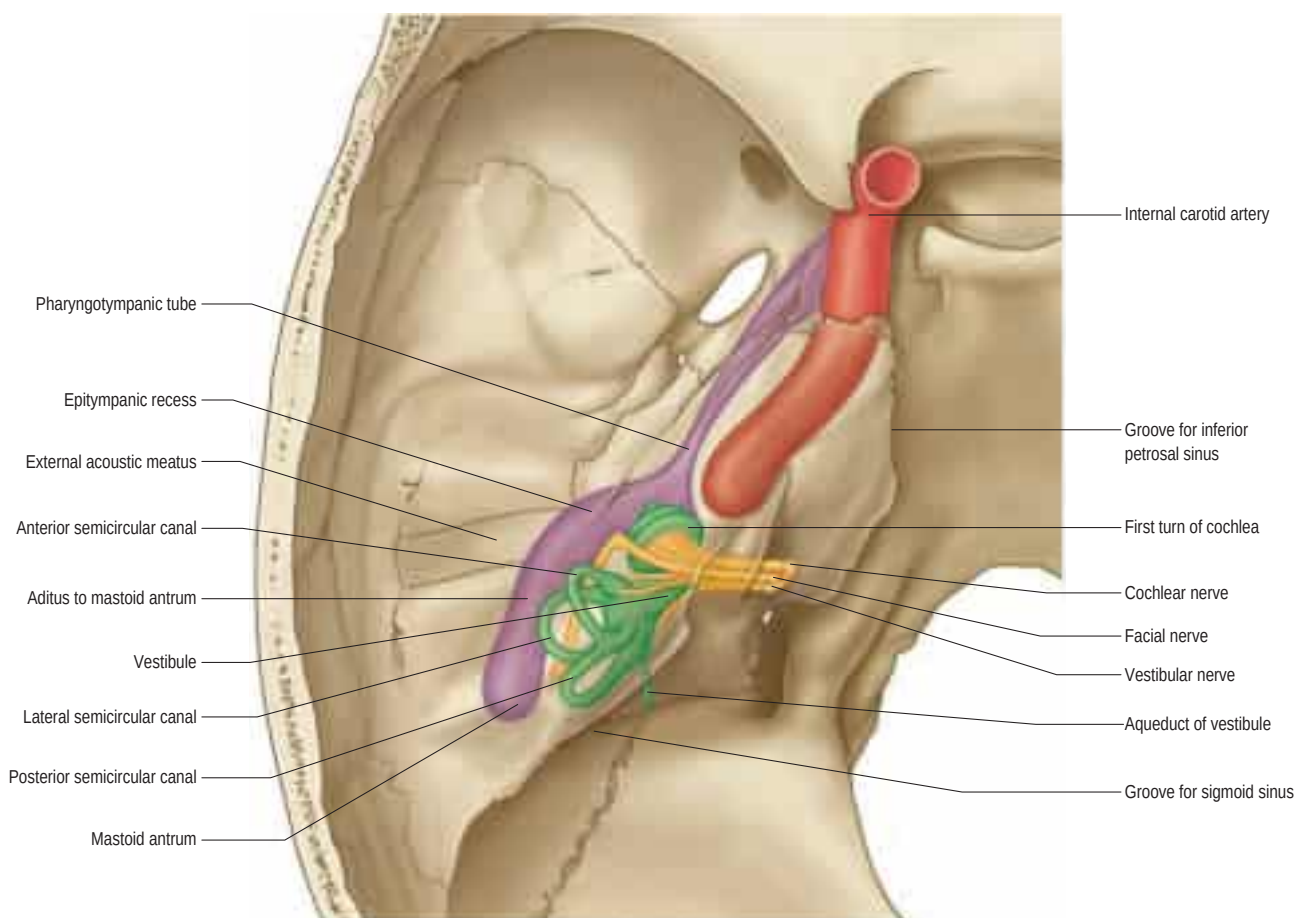


Fig. 37.7 The left auditory apparatus as if viewed through a semi-transparent temporal bone. Note the first genu in the facial nerve at the site of the geniculate ganglion.



Fig. 37.8 A bone-windowed, coronal computed tomography (CT) scan of the external ear canal and middle ear showing the curvature of the floor and roof, and the obliquity of the tympanic membrane (arrow) at the medial end of the external acoustic meatus. (Courtesy of Dr Steven Connor.)

for stapedectomy, ossiculoplasty, myringoplasty and the removal of small middle-ear tumours.

If the external acoustic meatus is too narrow to allow adequate visualization of the middle ear, or if access is required to the mastoid aditus and antrum, it is necessary to displace the superficial soft tissues. There are two main external approaches to the middle ear: the endaural approach and the postauricular approach.

The endaural approach involves making an incision in the notch between the tragus and the helix. This is carried down to expose the lower margin of temporalis muscle and the bone of the external acoustic meatus. The cartilaginous meatus is separated from the bony meatus and reflected laterally as a conchomeatal flap. The bony meatus can then be widened by drilling away bone (canalplasty). This gives more space to manipulate the delicate structures of the middle ear, as well as improving subsequent visualization of the tympanic membrane when the incision has healed.

The postauricular approach involves making an incision approximately 1 cm behind the postauricular skin crease and deepening the incision to the periosteum of the mastoid process, dividing the posterior auricular muscles on the way. Grafts can be harvested from the temporalis fascia. The periosteum is incised and elevated to expose the bony external acoustic meatus from behind. The skin over the junction of the bony and cartilaginous meatus is incised to allow the cartilage of the auricle and meatus to be swung forwards on its blood supply and so expose the bony meatus and mastoid process. Access can then be gained by drilling and elevating a tympanomeatal skin flap, as described for the endaural approach.

Temporalis fascia is the most popular tissue used as a free graft for repair of the tympanic membrane because it is easily obtainable. In recent years, tragal perichondrium has become a popular alternative; it has the additional advantage that the cartilage can also be harvested and used to reinforce the repair.

More extensive resections of the temporal bone are undertaken using extended pre- or postauricular incisions into the temporal region and neck. The blood supply of the pinna is sufficient to maintain viability despite significant elevation and undermining.

MIDDLE EAR

The middle ear is an irregular, laterally compressed space in the petrous part of the temporal bone. It is lined with mucous membrane and filled with air, which reaches it from the nasopharynx via the pharyngotympanic tube (see Figs 37.6–37.7; Fig. 37.9). The middle ear contains three small bones – the malleus, incus and stapes, collectively called the auditory ossicles – which form an articulated chain connecting the lateral and medial walls of the cavity, and which transmit the vibrations of the tympanic membrane across the cavity to the cochlea.

The essential function of the middle ear is to transfer energy efficiently from relatively weak vibrations in the elastic, compressible air in the external acoustic meatus to the incompressible fluid around the delicate receptors in the cochlea. Mechanical coupling between the two systems must match their resistance to deformation or 'flow', i.e. their impedance, as closely as possible. Aerial waves of low amplitude and low force per unit area arrive at the tympanic membrane, which has



Tensor
tympani

Malleus



Incudo-
stapedial joint

Incus



Incudo-
stapedial joint

Fig. 37.9, cont'd C, D, E, Coronal CT scans showing details of the ossicular mass. **C,** Malleus and tensor tympani. **D** and **E,** Incus and incudostapedial joint.

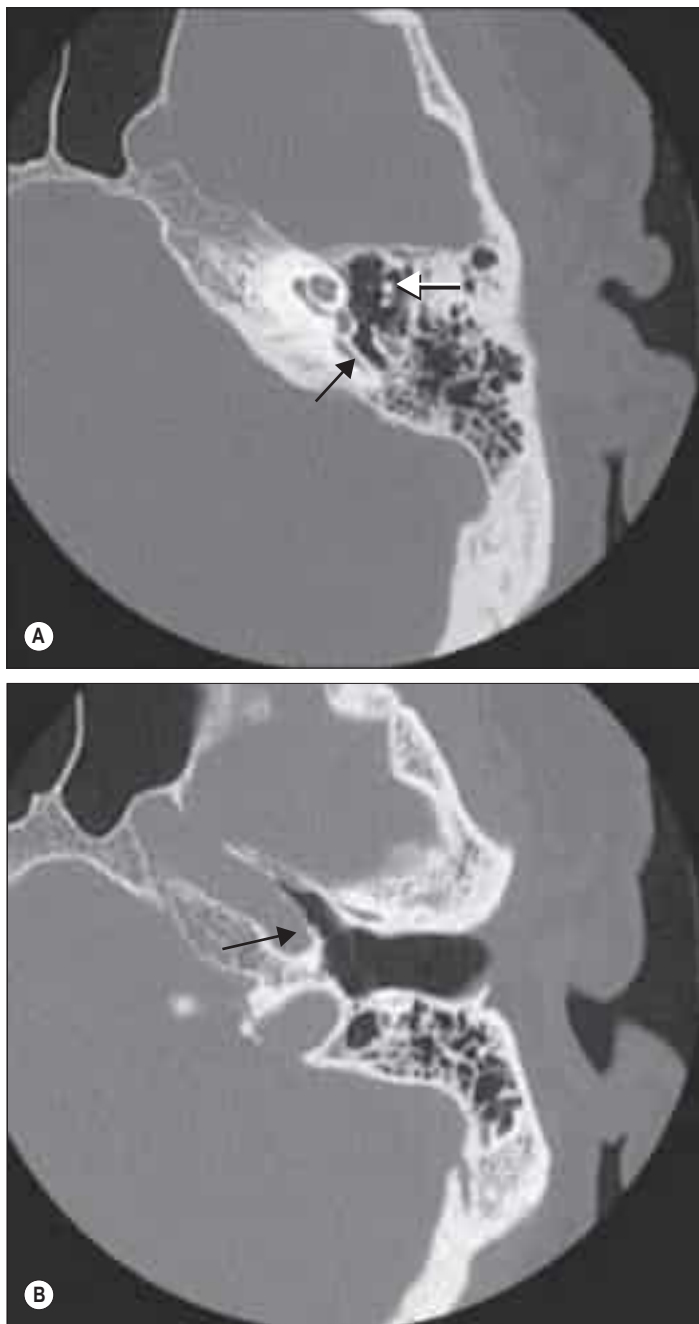


Fig. 37.9 **A**, An axial CT scan at the level of the ossicular mass (white arrow) in the epitympanum, showing the mastoid antrum and its relationship to the lateral (horizontal) semicircular canal (black arrow). **B**, An axial CT scan at the level of the entrance of the pharyngotympanic tube, showing its relationship to the internal carotid artery (black arrow). (A, B, Courtesy of Dr Steven Connor.) *Continued online*

15–20 times the area of the stapedial footplate that contacts the perilymph in the inner ear; the force per unit area generated by the footplate is increased by a similar amount, while the amplitude of vibration is almost unchanged.

Protective mechanisms incorporated into the design of the middle ear include the presence of the pharyngotympanic tube (to equalize pressure on both sides of the delicate tympanic membrane); the shape of the articulations between the ossicles; and the reflex contractions of stapedius and tensor tympani in response to sounds of fairly high intensity (preventing damage caused by sudden or excessive excursions of the ossicles).

The space within the middle ear can be subdivided into three parts. These are the mesotympanum or tympanic cavity proper, which is opposite the tympanic membrane; the epitympanum or attic, which is above the level of the membrane, and contains the head of the malleus and the body and short process of the incus; and the hypotympanum, which is in the floor of the cavity between the jugular bulb and the lower margin of the tympanic membrane. The vertical and anteroposterior diameters of the mesotympanum and hypotympanum are each approximately 15 mm; the transverse diameter is 6 mm superiorly and 4 mm inferiorly, narrowing to 2 mm opposite the umbo. The cavity is

bounded laterally by the tympanic membrane and medially by the lateral wall of the internal ear, the promontory. It communicates posteriorly with the mastoid antrum and the mastoid air cells, and anteriorly with the nasopharynx via the pharyngotympanic tube (see Figs 37.6–37.7).

The tympanic cavity and mastoid antrum, auditory ossicles and structures of the internal ear are all almost fully developed at birth and subsequently alter little; almost all of the volume changes are due to expansion of the epitympanic space (Osborn et al 2011). In the fetus, the cavity contains a gelatinous tissue that has practically disappeared by birth, when it is filled by a fluid that is absorbed when air enters via the pharyngotympanic tube. The tympanic cavity is a common site of infection in childhood.

BOUNDARIES OF THE TYMPANIC CAVITY

The tympanic cavity has a roof, a floor and lateral, medial, posterior and anterior walls.

Roof

A thin plate of compact bone, the tegmen tympani, separates the cranial and tympanic cavities, and forms much of the anterior surface of the petrous temporal bone. It is prolonged posteriorly as the roof of the mastoid antrum and anteriorly it covers the canal for tensor tympani. In youth, the unossified petrosquamosal suture may allow the spread of infection from the tympanic cavity to the meninges. In adults, veins from the tympanic cavity traverse this suture to reach the superior petrosal or petrosquamous sinus and thus may also transmit infection to these structures through a process of thrombophlebitis.

Longitudinal fractures of the middle cranial fossa almost always involve the tympanic roof, accompanied by dislocation of the ossicular chain, rupture of the tympanic membrane, or a fractured roof of the osseous external acoustic meatus, which can be seen as a notch on otoscopy. Such injuries usually cause bleeding from the ear, with escape of cerebrospinal fluid if the dura mater has been torn (CSF otorrhoea).

Floor

The floor of the tympanic cavity is a narrow, thin, convex plate of bone that separates the cavity from the superior bulb of the internal jugular vein. The bone may be patchily deficient, in which case the tympanic cavity and the vein are separated only by mucous membrane and fibrous tissue. Alternatively, the floor is sometimes thick and may contain some accessory mastoid air cells. A small aperture for the tympanic branch of the glossopharyngeal nerve lies near the medial wall.

Lateral wall

The lateral wall consists mainly of the tympanic membrane but also contains the ring of bone to which the membrane is attached (see Fig. 37.15A). The lateral epitympanic bony wall is wedge-shaped in section and its sharp inferior portion is known as the outer attic wall or scutum. This part is easily eroded or blunted by cholesteatoma, a feature easily detected on computed tomography (CT) scans (see Fig. 37.8). There is a deficiency or notch in the upper part of this ring, close to which are the small openings of the anterior and posterior canaliculi for the chorda tympani and the petrotympanic fissure. The posterior canaliculus for the chorda tympani is situated in the angle between the posterior and lateral walls of the tympanic cavity just behind the tympanic membrane, at a variable position approximately level with the upper end of the handle of the malleus. This variable position or entrance into the tympanic cavity should be kept in mind when mobilizing the chorda tympani in stapes surgery. It may not reflect inferiorly to the extent desired and can be easily damaged in the process. Within the mastoid, the chorda tympani passes into a minute canal that descends in front of the facial canal and ends in it about 6 mm above the stylomastoid foramen. This canal and the chorda tympani should not be mistaken for the main trunk of the facial nerve when skeletonizing the Fallopian canal during surgery. The canaliculus transmits the chorda tympani and a branch of the stylomastoid artery to the tympanic cavity. The chorda tympani leaves the tympanic cavity through the anterior canaliculus, which opens at the medial end of the petrotympanic fissure.

The petrotympanic fissure is a mere slit approximately 2 mm in length, which opens just above and in front of the ring of bone to which

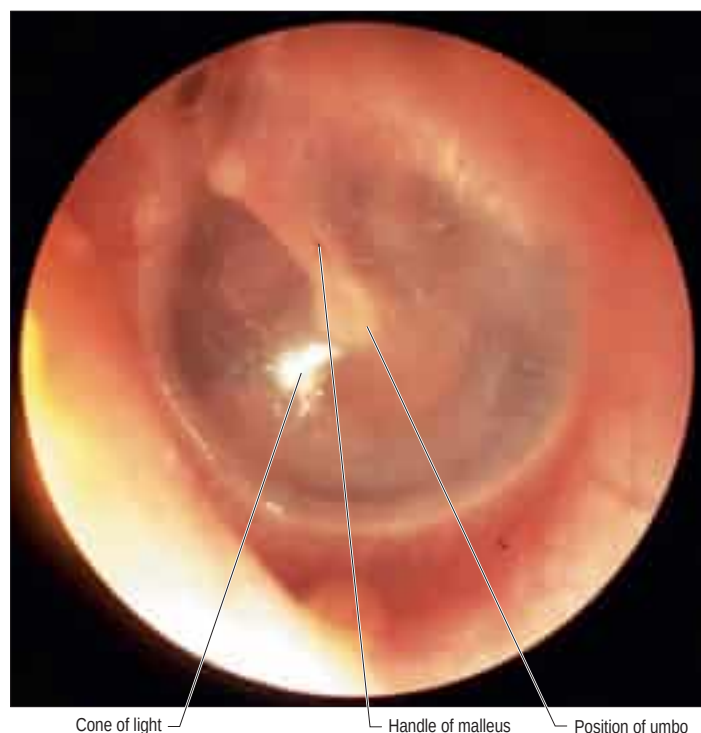


Fig. 37.10 An auroscopic view of the left tympanic membrane. Note that a bright cone of light is seen in the anteroinferior quadrant of the membrane when it is illuminated. (Courtesy of Mr Simon A Hickey.)

the tympanic membrane is attached. It contains the anterior process and anterior ligament of the malleus, and transmits the anterior tympanic branch of the maxillary artery to the tympanic cavity.

Tympanic membrane

The tympanic membrane separates the tympanic cavity from the external acoustic meatus (Fig. 37.10; see Figs 37.6, 37.15). It is thin, semi-transparent and almost oval, though somewhat broader above than below. It lies obliquely, at an angle of approximately 55° with the meatal floor in adults. Its longest, anteroinferior diameter is 9–10 mm, and its shortest is 8–9 mm. Most of its circumference is a thickened fibrocartilaginous ring or annulus, which is attached to the tympanic sulcus at the medial end of the meatus. The annulus contains radially orientated smooth muscle cells in several locations that possibly play a role in controlling blood flow or maintaining tension (Henson et al 2005). The sulcus is deficient superiorly, i.e. it is notched. Two bands, the anterior and posterior malleolar folds, pass from the ends of this notch to the lateral process of the malleus. The small triangular part of the membrane, the pars flaccida, lies above these folds and is lax and thin. The major part of the tympanic membrane, the pars tensa, is taut. The handle of the malleus is firmly attached to the internal surface of the tympanic membrane as far as its centre, which projects towards the tympanic cavity. The inner surface of the membrane is thus convex and the point of greatest convexity is termed the umbo. Although the membrane as a whole is convex on its inner surface, its radiating fibres are curved with their concavities directed inwards.

Microstructure

Histologically, the tympanic membrane is composed of an outer cuticular layer, an intermediate fibrous layer and an inner mucous layer.

The cuticular stratum is continuous with the thin skin of the meatus. It is keratinized, stratified squamous in type, devoid of dermal papillae and hairless. Its subepithelial tissue is vascularized and may develop a few peripheral papillae. Ultrastructurally, it is typically 10 cells thick and has two zones: a superficial layer of non-nucleated squames, and a deep zone that resembles the epidermal prickle cell layer (stratum spinosum). There are numerous desmosomes between cells, the deepest of which lie on a continuous basal lamina, but lack epithelial pegs and hemidesmosomes. The cells of this stratum have a propensity for lateral migration and differentiation not shared with any other stratified squamous epithelia in the body.

The fibrous stratum consists of an external layer of radiating fibres that diverge from the handle of the malleus, and a deep layer of circular fibres, which are plentiful peripherally but sparse and scattered cen-

trally. Ultrastructurally, the filaments are 10 nm in diameter and are linked at 25 nm intervals. They have a distinctive amino acid composition, and may consist of a protein peculiar to the tympanic membrane. Small groups of collagen fibrils appear at 11 weeks *in utero*, interspersed with small bundles of elastin microfibrils. Older specimens contain more typically cross-banded collagen fibrils and an amorphous elastin component. The fibrous stratum is replaced by loose connective tissue in the pars flaccida.

The mucous stratum is a part of the mucosa of the tympanic cavity and is thickest near the upper part of the membrane. It consists of a single layer of very flat cells, with overlapping interdigitating boundaries and desmosomes and tight junctions between adjacent cells. The cytoplasm contains only a few organelles; the luminal surfaces of these apparently metabolically inert cells have a few irregular microvilli and are covered by an amorphous electron-dense material. There are no ciliated columnar cells.

Innervation

The tympanic membrane is mainly innervated by the auriculotemporal nerve and appears to perceive only pain. There is a minor, inconstant and overlapping sensory supply from the facial, glossopharyngeal and vagus nerves.

The auricular branch of the vagus arises from the superior vagal ganglion and is joined soon after by a ramus from the inferior ganglion of the glossopharyngeal nerve. It passes behind the internal jugular vein and enters the mastoid canaliculus on the lateral wall of the jugular fossa. It traverses the temporal bone and crosses the facial canal about 4 mm above the stylomastoid foramen. At this point, it supplies an ascending branch to the facial nerve. Fibres of the nervus intermedius may pass to the auricular branch of the vagus here, which may explain the cutaneous vesiculation that sometimes accompanies geniculate herpes. The auricular branch then traverses the tympanomastoid fissure and divides into two rami. One ramus joins the posterior auricular nerve and the other is distributed to the skin of part of the cranial surface of the auricle, the posterior wall and floor of the external acoustic meatus, and to the adjoining part of the outer surface of the tympanic membrane. The auricular branch therefore contains somatic afferent nerve fibres, which probably terminate in the spinal trigeminal nucleus. Stimulation of the vagus nerve, e.g. in syringing the ear, can cause a reflex bradycardic reaction on heart rate. From a practical point of view, the tympanic membrane can be anaesthetized for minor procedures like myringotomy by the application of topical anaesthetic agents. For more complicated surgical procedures, e.g. tympanoplasty, it is possible to provide sufficient anaesthesia by local infiltration at the level of the bony and cartilaginous external acoustic meatus.

Otitis media

It is assumed that acute otitis media usually arises as a result of ascending infection from the nasopharynx via the pharyngotympanic tube to the middle ear cleft. From there, it may extend to the mastoid aditus and antrum. Swelling secondary to the infection may result in the closure of both exits from the middle ear, i.e. the pharyngotympanic tube and the aditus, with subsequent accumulation of pus under pressure, which causes lateral bulging and inflammation of the tympanic membrane. The latter may burst, releasing mucopurulent discharge into the external acoustic meatus, which results in a release of the pressure in the middle ear and a diminution in the levels of pain. After a brief period, the discharge dries up and, for the most part, the resultant perforation of the tympanic membrane heals. Normal ventilation and drainage of mucus from the middle ear are restored once the swelling and function of the pharyngotympanic tube resolve. On occasion, the process will fail to produce a perforation of the tympanic membrane and the inflammatory exudates will not drain. The immune defence system sterilizes the exudates of organisms, resulting in a sterile mucoid effusion, otitis media with effusion or glue ear (see below). The effusion may cause protracted deafness because its relatively incompressible nature prevents free vibration of the tympanic membrane (Bluestone and Klein 2002).

Myringoplasty

Persistent perforation of the tympanic membrane caused by infection or trauma leads to hearing impairment and predisposes to continuing infection as a result of contamination with organisms from the external acoustic meatus. This condition is known as chronic suppurative otitis media of the tubotympanic type. Myringoplasty is a surgical procedure that uses a connective tissue scaffold or graft to support healing of the perforation. The most common technique involves the elevation of the tympanic annulus and the placement of a piece of fibrous connective tissue, e.g. part of the fibrous deep fascia that invests the lateral surface of temporalis or the perichondrium of the tragal cartilage, on to the

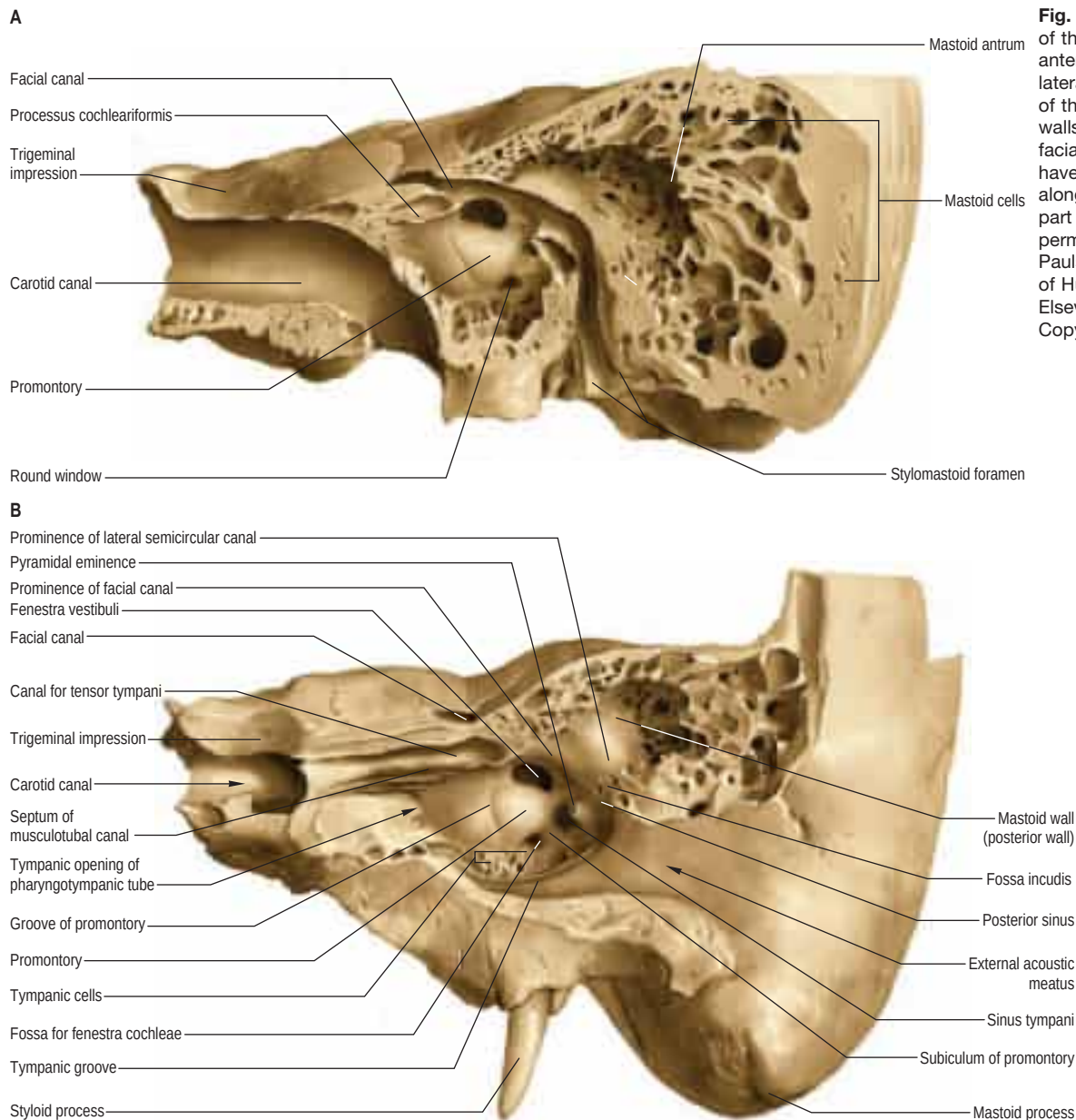


Fig. 37.11 The medial wall of the left tympanic cavity, anterolateral aspect. **A**, The lateral wall and adjacent parts of the anterior and superior walls have been removed; the facial canal and carotid canal have been opened. **B**, A section along the axis of the petrous part of the temporal bone. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

undersurface of the tympanic membrane to close the perforation. The healed edges of the perforation are stripped of epithelium to encourage healing and scar formation. The fibrous tissue supports the healing tympanic membrane and may, in part, be incorporated into the repair. Once the perforation is healed, the vibratory function of the tympanic membrane is usually restored to normal.

Medial wall

The medial wall of the tympanic cavity is also the lateral boundary of the internal ear. Its features are the promontory, fenestra vestibuli (fenestra ovalis, oval window), fenestra cochleae (fenestra rotunda, round window) and the facial prominence (Fig. 37.11).

The promontory is a rounded prominence furrowed by small grooves that lodge the nerves of the tympanic plexus. It lies over the lateral projection of the basal turn of the cochlea. A minute spicule of bone frequently connects the promontory to the pyramidal eminence of the posterior wall. The apex of the cochlea lies near the medial wall of the tympanic cavity, anterior to the promontory. A depression behind the promontory is known as the sinus tympani.

The fenestra vestibuli is a kidney-shaped opening situated above and behind the promontory, and leading from the tympanic cavity to the vestibule of the inner ear. Its long diameter is horizontal and its convex border is directed upwards. It is occupied by the base of the stapes, the footplate; the circumference of the footplate is attached to the margin of the fenestra by an annular ligament.

The fenestra cochleae is situated below and a little behind the fenestra vestibuli, from which it is separated by a posterior extension of the promontory, called the subiculum. Occasionally, another ridge

of bone, the ponticulus, leaves the promontory above the subiculum and runs to the pyramid on the posterior wall of the cavity. The fenestra cochleae lies completely under the overhanging edge of the promontory in a deep hollow or niche, and is placed very obliquely. In dried specimens, it opens anterosuperiorly from the tympanic cavity into the scala tympani of the cochlea, but in life it is closed by the secondary tympanic membrane. This is somewhat concave towards the tympanic cavity and convex towards the cochlea, and is bent so that its posterosuperior third forms an angle with its anteroinferior two-thirds. The membrane is composed of an external layer derived from the tympanic mucosa; an internal layer, derived from the cochlear lining membrane; and an intermediate, fibrous layer.

The prominence of the facial nerve canal indicates the position of the upper part of the bony facial canal (Fallopian canal), which contains the facial nerve. The canal crosses the medial tympanic wall from the cochleariform process anteriorly, runs just above the fenestra vestibuli, and then curves down into the posterior wall of the cavity. Its lateral wall may be partly deficient.

Posterior wall

The posterior wall of the tympanic cavity is wider above than below. Its main features are the aditus to the mastoid antrum, the pyramid and the fossa incudis (see Fig. 37.11).

The aditus to the mastoid antrum is a large irregular aperture that leads back from the epitympanic recess into the upper part of the mastoid antrum. A rounded eminence on the medial wall of the aditus, above and behind the prominence of the facial nerve canal, corresponds to the position of the lateral semicircular canal.

The pyramidal eminence is situated just behind the fenestra vestibuli and in front of the vertical part of the facial nerve canal. It is hollow and contains the stapedius muscle. Its summit projects towards the fenestra vestibuli and is pierced by a small aperture that transmits the tendon of stapedius. The cavity in the pyramidal eminence is prolonged down and back in front of the facial nerve canal; it communicates with the canal by an aperture through which a small branch of the facial nerve passes to stapedius.

The fossa incudis is a small depression in the lower and posterior part of the epitympanic recess. It contains the short process of the incus, which is fixed to the fossa by ligamentous fibres.

Mastoid antrum

The mastoid antrum is an air sinus in the petrous part of the temporal bone. Its topographical relations are of considerable surgical importance. The aditus to the mastoid antrum, which leads back from the epitympanic recess, opens in the upper part of its anterior wall. The lateral semicircular canal lies medial to the aditus. The descending part of the facial nerve canal is anteroinferior. The medial wall is related to the posterior semicircular canal. The sigmoid sinus lies some distance posteriorly; the distance can be extremely variable and is dependent on the degree of pneumatization of the mastoid. The roof is formed by the tegmen tympani, and so the antrum lies below the middle cranial fossa and the temporal lobe of the brain. The floor has several openings that communicate with the mastoid air cells. The lateral wall, which offers the usual surgical approach to the cavity, is formed by the postmeatal process of the squamous part of the temporal bone. This is only 2 mm thick at birth but increases at an average rate of 1 mm a year, attaining a final thickness of 12–15 mm. In adults, the lateral wall of the antrum corresponds to the suprameatal triangle (Macewen's triangle) on the outer surface of the skull. This is palpable through the cymba conchae. The superior side of the triangle, the supramastoid crest, is level with the floor of the middle cranial fossa; the anteroinferior side, which forms the posterosuperior margin of the external acoustic meatus, indicates approximately the position of the descending part of the facial nerve canal; and the posterior side, formed by a posterior vertical tangent to the posterior margin of the external acoustic meatus, is anterior to the sigmoid sinus.

The adult capacity of the mastoid antrum is variable, but on average is 1 ml, with a general diameter of 10 mm. Unlike the other air sinuses in the skull, it is present at birth, and indeed is then almost adult in size, although it is at a higher level relative to the external acoustic meatus than it is in adults. In the very young, the thinness of the lateral antral wall and the absence or underdevelopment of the mastoid process mean that the stylomastoid foramen and emerging facial nerve are very superficially situated.

Mastoid air cells

Though the mastoid antrum is well developed at birth, the mastoid air cells are merely minute antral diverticula at this stage. As the mastoid develops in the second year, the air cells gradually extend into it and by the fourth year they are well formed, although their greatest growth occurs at puberty. They vary considerably in number, form and size. Usually, they interconnect and are lined by a mucosa with squamous non-ciliated epithelium, continuous with that in the mastoid antrum and tympanic cavity. They may fill the mastoid process, even to its tip, and some may be separated from the sigmoid sinus and posterior cranial fossa only by extremely thin bone, which is occasionally deficient (see Fig. 37.11). Some may lie superficial to, or even behind, the sigmoid sinus, and others may be present in the posterior wall of the descending part of the facial nerve canal. Those in the squamous part of the temporal bone may be separated from deeper cells in the petrous part by a plate of bone in the line of the squamomastoid suture (Körner's septum). Sometimes, they extend only minimally into the mastoid process, in which case the process consists largely of dense bone or trabecular bone containing bone marrow. Varieties of the mastoid process are recognized. The three types most commonly described are pneumatized (with many air cells); sclerotic or diploic (with few or no air cells); and mixed (contain both air cells and bone marrow).

The mastoid process may sometimes have no air cells at all. Alternatively, air cells may extend beyond the mastoid process into the squamous part of the temporal bone above the supramastoid crest; into the posterior root of the zygomatic process of the temporal bone; into the osseous roof of the external acoustic meatus just below the middle cranial fossa; or into the floor of the tympanic cavity very close to the superior jugular bulb. Rarely, a few may excavate the jugular process of the occipital bone. An important group may extend medially into the petrous part of the temporal bone, even to its apex, and are related to the pharyngotympanic tube, carotid canal, labyrinth and abducens

nerve. Some investigators maintain that these are not continuous with the mastoid cells but grow independently from the tympanic cavity. All of these extensions of the mastoid air cells are pathologically important since infection may spread to the structures around them. (For further reading on the mastoid air cells and other pneumatized regions of the temporal bone, see Allam (1969).)

Innervation The mastoid air cells are innervated by a meningeal branch of the mandibular division of the trigeminal nerve.

Mastoiditis

Mastoiditis is a potentially dangerous, life-threatening condition that develops as a result of the spread of bacterial infection from the tympanic cavity via the aditus to the mastoid antrum and associated mastoid air cells. Occasionally, the infection may spread through the tegmen tympani to the dura mater of the middle cranial fossae, to cause meningitis or a temporal lobe abscess. Similar spread may be seen into the posterior cranial fossa and cerebellum.

Infection may spread laterally through the cortical bone of the lateral aspect of the mastoid process to form a subperiosteal postauricular abscess (Bezold's abscess), or through the cortical bone of the tip of the mastoid process to the attachment of the posterior belly of digastric and sternocleidomastoid, which stimulates painful muscular contraction and torticollis.

Anterior wall

The inferior, larger area of the anterior wall of the tympanic cavity is narrowed by the approximation of the medial and lateral walls of the cavity (see Fig. 37.6). It is a thin lamina and forms the posterior wall of the carotid canal. It is perforated by the superior and inferior carotico-tympanic nerves and the tympanic branch or branches of the internal carotid. The canals for tensor tympani and the osseous part of the pharyngotympanic tube open above it, the canal for tensor tympani being superior to that for the pharyngotympanic tube. Both canals incline downwards and anteromedially, to open in the angle between the squamous and petrous parts of the temporal bone, and are separated by a thin, osseous septum. The canal for tensor tympani and the bony septum runs posterolaterally on the medial tympanic wall, and ends immediately above the fenestra vestibuli. Here, the posterior end of the septum is curved laterally to form a pulley, the processus cochleariformis (cochleariform process), which is a surgical landmark for the identification of the geniculate ganglion of the facial nerve. The tendon of tensor tympani turns laterally over the pulley before attaching to the upper part of the handle of the malleus.

Pharyngotympanic tube blockage in children

The pharyngotympanic tube serves to ventilate the middle ear, exchanging nasopharyngeal air with the air in the middle ear, which has been altered in its composition via transmucosal gas exchange with the haemoglobin in the blood vessels of the mucosa. The tube also carries mucus from the middle ear cleft to the nasopharynx as a result of ciliary transport.

In children, the pharyngotympanic tube is relatively narrow. It is prone to obstruction when the mucosa swells in response to infection or allergic challenge; obstruction results in a relative vacuum being created in the middle ear secondary to transmucosal gas exchange, and this in turn promotes mucosal secretion and the formation of a middle ear effusion. Because of the collapsibility of the pharyngotympanic tube, the vacuum thus created can overcome the distending effect of the muscles of the tube and 'lock' the tube shut. The resultant persistent middle ear effusion, otitis media with effusion (glue ear), can cause hearing loss by splinting the tympanic membrane and impeding its vibration. It can also provide an ideal environment for the proliferation of bacteria, with the result that an acute otitis media may develop (see above). It is possible to relieve the vacuum and unlock the tube, and then remove the effusion by myringotomy, i.e. by surgically creating a hole in the tympanic membrane. This hole will generally heal rapidly and it is common practice to insert a flanged ventilation tube (a grommet or tympanostomy tube) to keep the hole open. Migration of the outer squamous layer of the tympanic membrane eventually displaces the tube and the myringotomy heals.

AUDITORY OSSICLES

A chain of three mobile ossicles – the malleus, incus and stapes – transfers sound waves across the tympanic cavity from the tympanic

EXTERNAL AND MIDDLE EAR

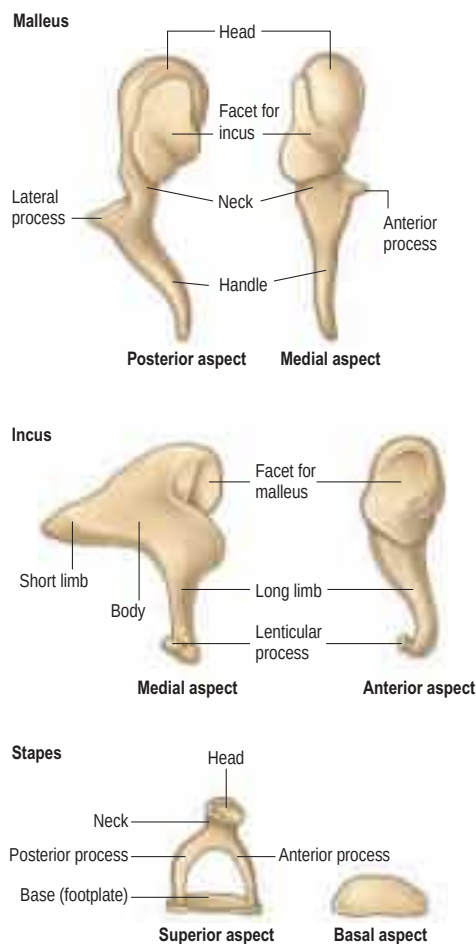


Fig. 37.12 The left auditory ossicles.

membrane to the fenestra vestibuli. The malleus is attached to the tympanic membrane, and the base (footplate) of the stapes is attached to the rim of the fenestra vestibuli. The incus is suspended between them and articulates with both bones.

Malleus

The malleus is the largest of the ossicles and is shaped somewhat like a mallet (**Fig. 37.12**). It is 8–9 mm long and has a head, neck, handle (manubrium) and anterior and lateral processes. The head is the large upper end of the bone and is situated in the epitympanic recess. It is ovoid in shape, articulates posteriorly with the incus, and is covered elsewhere by mucosa. The cartilaginous articular facet for the incus is narrowed near its middle and consists of a larger upper part and a smaller lower part, orientated almost at right angles to each other. Opposite the constriction, the lower margin of the facet projects in the form of a process, the spur of the malleus. The neck is the narrowed part below the head, and inferior to this is an enlargement from which the anterior and lateral processes project.

The handle of the malleus is connected by its lateral margin to the tympanic membrane (see **Figs 37.6, 37.10, 37.15A**). It is directed downwards, medially and backwards. It decreases in size towards its free end, which is curved slightly forwards and is flattened transversely. Near the upper end of its medial surface there is a slight projection to which the tendon of tensor tympani is attached. The anterior process is a delicate bony spicule, directed forwards from the enlargement below the neck, and connected to the petrotympanic fissure by ligamentous fibres. In fetal life, it is the longest process of the malleus and is continuous in front with Meckel's cartilage. The lateral process is a conical projection from the root of the handle of the malleus. It is directed laterally, and is attached to the upper part of the tympanic membrane and, via the anterior and posterior malleolar folds, to the sides of the notch in the upper part of the tympanic sulcus.

Ossification

The cartilaginous precursor of the malleus originates as part of the dorsal end of Meckel's cartilage. With the exception of its anterior process, the malleus ossifies from a single endochondral centre that appears near the future neck of the bone in the fourth month *in utero*. The anterior process ossifies separately in dense connective tissue and joins the rest of the bone at about the sixth month *in utero*.

Incus

The incus is shaped less like an anvil, from which it is named, than a premolar tooth with its two diverging roots. It has a body and two processes (see **Fig. 37.12**). The body is somewhat cubical but laterally compressed. On its anterior surface, it has a saddle-shaped facet for articulation with the head of the malleus. The long process, rather more than half the length of the handle of the malleus, descends almost vertically, behind and parallel to the handle. Its lower end bends medially and ends in a rounded lenticular process, the medial surface of which is covered with cartilage and articulates with the head of the stapes. The short process, somewhat conical, projects backwards and is attached by ligamentous fibres to the fossa incudis in the lower and posterior part of the epitympanic recess. The presence of a single nutrient vessel within the long process of the incus and the absence of any collateral circulation render this segment of the bone susceptible to aseptic necrosis following middle ear infections.

Ossification

The incus has a cartilaginous precursor continuous with the dorsal extremity of Meckel's cartilage. Ossification often spreads from a single centre in the upper part of its long process in the fourth fetal month; the lenticular process may have a separate centre.

Stapes

The stapes is also known as the stirrup. It has a head, neck, two limbs (processes or crura) and a base (footplate) (see **Fig. 37.12**). The head (caput) is directed laterally and has a small cartilaginous facet for articulation with the lenticular process of the incus. The neck is the constricted part supporting the head, and the tendon of stapedius is attached to its posterior surface. The processes diverge from the neck and are connected at their ends by a flattened oval plate, the base, which forms the footplate of the stapes. The footplate is attached to the margin of the fenestra vestibuli by a ring of fibres (the annular ligament). The anterior process is shorter, thinner and less curved than the posterior.

Ossification

The stapes is preformed in the perforated dorsal moiety of the hyoid arch cartilage of the fetus. Ossification starts from a single endochondral centre, which appears in the base in the fourth month *in utero* and then gradually spreads through the limbs of the stapes to reach the head.

At birth, the auditory ossicles have achieved their full adult size and configuration. They increase in density during the first years of life as marrow cavities are replaced with endosteal bone and fine trabeculae (**Isacson 2014**).

Ossicular ligaments

The ossicles are connected to the tympanic walls by ligaments (see **Fig. 37.6B**): three for the malleus and one each for the incus and stapes. Some are mere mucosal folds that carry blood vessels and nerves to and from the ossicles and their articulations, and others contain a central, strong band of collagen fibres.

The anterior ligament of the malleus stretches from the neck of the malleus, just above the anterior process, to the anterior wall of the tympanic cavity near the petrotympanic fissure. Some of its collagen fibres traverse this fissure to reach the spine of the sphenoid, and others continue into the sphenomandibular ligament. The latter, like the anterior malleolar ligament, is derived from the perichondrial sheath of Meckel's cartilage. The anterior malleolar ligament may contain muscle fibres, called laxator tympani or musculus externus mallei. The lateral ligament of the malleus is a triangular band that stretches from the posterior part of the border of the tympanic incisura to the head of the malleus. The superior ligament of the malleus connects the head of the malleus to the roof of the epitympanic recess.

The posterior ligament of the incus connects the end of its short process to the fossa incudis. The superior ligament of the incus is little more than a mucosal fold passing from the body of the incus to the roof of the epitympanic recess.

The vestibular surface and rim of the stapedial base are covered with hyaline cartilage. The cartilage encircling the base is attached to the margin of the fenestra vestibuli by a ring of elastic fibres, the annular ligament of the base of the stapes. The posterior part of this ligament is much narrower than the anterior part; it acts as a kind of hinge on which the stapedial base moves when stapedius contracts and during acoustic oscillation.

Ossicular articulations

The ossicular articulations are typical synovial joints. The incudomalleolar joint is saddle-shaped and the incudostapedial joint is a ball and socket articulation. The articular surfaces are covered with articular cartilage, and each joint is enveloped by a capsule rich in elastic tissue and lined by synovial membrane. Versican, fibronectin, elastin and hyaluronan have been co-localized in the attachments of tensor tympani and stapedius to the malleus and stapes, respectively, and in the annular ligament of the incudostapedial joint in histological sections of adult middle ears (Kawase et al 2012). Co-localization between elastic fibres and hyaluronan in middle ear entheses looks to be a postnatal maturational event, possibly an adaptation to acoustic oscillation producing multidirectional mechanical stress (Takanashi et al 2013).

Movements of the auditory ossicles

The handle of the malleus faithfully follows all movements of the tympanic membrane. The malleus and incus rotate together around an axis that runs from the short process and posterior ligament of the incus to the anterior ligament of the malleus. When the tympanic membrane and handle of the malleus move inwards (medially), the long process of the incus moves in the same direction and pushes the stapedial footplate towards the labyrinth and the perilymph contained within the labyrinth. The movement of the perilymph causes a compensatory outward bulging of the secondary tympanic membrane. These events are reversed when the tympanic membrane moves outwards. If the movement of the tympanic membrane is considerable, however, the incus does not follow the full outward excursion of the malleus and merely glides on it at the incudomalleolar joint, so preventing a dislocation of the footplate of the stapes from the fenestra vestibuli. When the handle of the malleus is carried medially, the spur at the lower margin of the head of the malleus locks the incudomalleolar joint, and this necessitates an inward movement of the long process of the incus. The joint is unlocked again when the handle of the malleus is carried outwards. The three bones together act as a bent lever so that the stapedial footplate does not move in the fenestra vestibuli like a piston, but rocks on a fulcrum at its anteroinferior border, where the annular ligament is thick. The rocking movement around a vertical axis, which is like a swinging door, is said to happen only at moderate intensities of sound. With loud, low-pitched sounds, the axis becomes horizontal, and the upper and lower margins of the stapedial footplate oscillate in opposite directions around this central axis, thus preventing excessive displacement of the perilymph.

Otosclerosis, stapedectomy and stapedotomy

Otosclerosis is a localized disease of the bone derived from the embryonic otic capsule in which lamellar bone is replaced by woven bone of greater thickness and vascularity. It may develop spontaneously but is often seen in pedigrees; the inherited form is associated with mutations of the *RELN* gene. The position of the focus of new bone formation determines its effect on the function of the ear. When new bone develops around the footplate of the stapes, it may fix the footplate to the margin of the fenestra vestibuli and prevent it from moving. This impedes the passage of vibrations of the tympanic membrane passing through the ossicular chain to the inner ear, producing a conductive hearing loss. Otosclerosis affecting other parts of the otic capsule is thought to cause a sensorineural element to the overall hearing loss.

Stapedectomy is a surgical procedure designed to bypass the fixation of the stapes footplate caused by otosclerosis or congenital fixation. The tympanic membrane is temporarily elevated for access to the middle ear and, under microscopic control, the incudostapedial joint is disarticulated. The limbs of the stapes and stapedius are then both divided and the superstructure of the stapes removed. A small hole (stapedotomy) is then made in the fixed footplate of the stapes using a microdrill, reamer or laser to expose the fluids of the inner ear. A small graft of connective tissue is used to seal the hole with a flexible membrane. A piston, usually made of Teflon or titanium incorporating a wire made of stainless steel, platinum or titanium, is crimped on to the long process of the incus and placed in the perforation in the stapes footplate. The tympanic membrane is then returned. The connection between the tympanic membrane and the inner ear is thus reconstituted and hearing restored.

MUSCLES

There are two intratympanic muscles: tensor tympani and stapedius (Anderson 1976).

Tensor tympani

Tensor tympani is a long, slender muscle that occupies the bony canal above the osseous part of the pharyngotympanic tube, from which it is separated by a thin bony septum (see Fig. 37.6). It arises from the cartilaginous part of the pharyngotympanic tube and the adjoining region of the greater wing of the sphenoid, as well as from its own canal. It passes back within its canal, and ends in a slim tendon which bends laterally round the pulley-like processus cochleariformis and attaches to the handle of the malleus, near its root (see Fig. 37.14).

Vascular supply Tensor tympani receives its arterial blood supply from the superior tympanic branch of the middle meningeal artery.

Innervation Tensor tympani is innervated by a branch of the nerve to medial pterygoid (a ramus of the mandibular division of the trigeminal nerve), which traverses the otic ganglion without interruption to reach the muscle.

Actions Tensor tympani draws the handle of the malleus medially, and so tenses the tympanic membrane and helps to damp sound vibrations; its action also pushes the base of the stapes more tightly into the fenestra vestibuli.

Stapedius

Stapedius arises from the wall of a conical cavity in the pyramidal eminence on the posterior wall of the tympanic cavity, and from its continuation anterior to the descending part of the facial nerve canal. Its minute tendon emerges from the orifice at the apex of the pyramid and passes forwards to attach to the posterior surface of the neck of the stapes. The muscle is of an asymmetric bipennate form. It contains numerous small motor units, each of only 6–9 muscle fibres; a few neuromuscular spindles exist near the myotendinous junction.

Vascular supply Stapedius receives its arterial blood supply from branches of the posterior auricular, anterior tympanic and middle meningeal arteries.

Innervation Stapedius is supplied by a branch of the facial nerve that is given off in the facial canal.

Actions Stapedius helps to damp down excessive sound vibrations. It opposes the action of tensor tympani (which pushes the stapes more tightly into the fenestra vestibuli). Paralysis of stapedius results in hyperacusis.

Stapedial and tensor tympani reflexes When noises are loud, and immediately before speaking, a reflex contraction of stapedius and tensor tympani takes place that helps damp down the movement of the ossicular chain before vibrations reach the internal ear. The afferent pathways involve the auditory component of cranial nerve VIII, and higher centres prior to speech. The efferent pathway involves the facial nerve (stapedius) and the mandibular nerve (tensor tympani).

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

A number of arteries supply the walls and contents of the tympanic cavity. The deep auricular, anterior tympanic and stylomastoid arteries are larger than the others.

The deep auricular branch of the first part of the maxillary artery often arises with the anterior tympanic artery. It ascends in the parotid gland behind the temporomandibular joint, pierces the cartilaginous or bony wall of the external acoustic meatus and supplies its cuticular lining, the exterior of the tympanic membrane and the temporomandibular joint.

The anterior tympanic branch of the first part of the maxillary artery ascends behind the temporomandibular joint and enters the tympanic cavity through the petrotympanic fissure. It ramifies on the interior of the tympanic membrane, and forms a vascular circle around it with the posterior tympanic branch of the stylomastoid artery. It also anastomoses with twigs of the artery of the pterygoid canal and caroticotympanic branches of the internal carotid artery in the mucosa of the tympanic cavity.

The stylomastoid branch of the occipital or posterior auricular arteries supplies the posterior part of the tympanic cavity and mastoid air cells. It also enters the stylomastoid foramen to supply the facial nerve and semicircular canals. In the young, its posterior tympanic branch forms a circular anastomosis with the anterior tympanic artery.

The smaller arteries supplying the tympanic cavity include the petrosal branch of the middle meningeal artery, which enters through the hiatus for the greater petrosal nerve; the superior tympanic branch of the middle meningeal artery, which traverses the canal for tensor tympani; an inferior tympanic branch from the ascending pharyngeal artery, which traverses the tympanic canaliculus together with the tympanic branch of the glossopharyngeal nerve, to supply the medial wall of the tympanic cavity; a branch from the artery of the pterygoid canal, which accompanies the pharyngotympanic tube; and a tympanic branch or branches from the internal carotid artery, which is given off in the carotid canal and perforates the thin anterior wall of the tympanic cavity.

The mastoid air cells and dura mater are also supplied by a mastoid branch from the occipital artery. It is small in size and sometimes absent. When present, it enters the cranial cavity via the mastoid foramen near the occipitomastoid suture.

In early fetal life, a stapedia artery traverses the stapes.

The veins from the tympanic cavity terminate in the pterygoid venous plexus and the superior petrosal sinus. A small group of veins runs medially from the mucosa of the mastoid antrum through the arch formed by the superior (anterior) semicircular canal, and emerges on to the posterior surface of the petrous temporal bone at the subarcuate fossa. These veins drain into the superior petrosal sinus and are the remains of the large subarcuate veins of childhood. They represent a potential route for the spread of infection from the mastoid antrum to the meninges.

Lymphatic vessels of the tympanic and antral mucosae drain to the parotid or upper deep cervical lymph nodes. Vessels draining the tympanic end of the pharyngotympanic tube probably end in the deep cervical nodes.

INNERVATION

The tympanic cavity contains the tympanic plexus and the facial nerve. Branches from the plexus and the facial nerve supply structures within the tympanic cavity but also leave the cavity to supply structures on the face.

Tympanic plexus

The nerves that constitute the tympanic plexus ramify on the surface of the promontory on the medial wall of the tympanic cavity. They are derived from the tympanic branch of the glossopharyngeal nerve and the caroticotympanic nerves (Fig. 37.13). The former arises from the

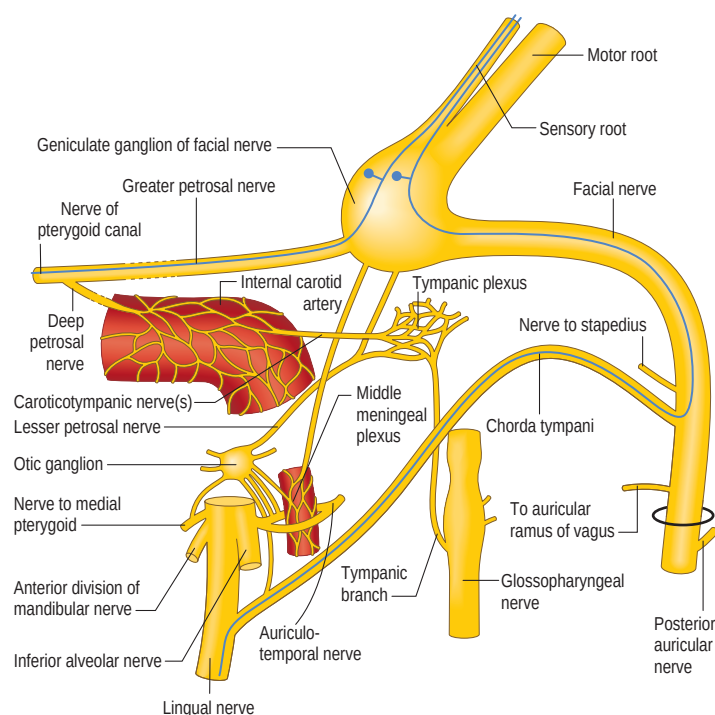


Fig. 37.13 The intracranial branches of the facial and glossopharyngeal nerves and the connections of the tympanic plexus and the otic ganglion. The course of the taste fibres from the mucous membrane of the palate and from the anterior presulcal part of the tongue is represented by the blue lines.

inferior ganglion of the glossopharyngeal nerve, and reaches the tympanic cavity via the tympanic canaliculus for the tympanic nerve. The superior and inferior caroticotympanic nerves are postganglionic sympathetic fibres that are derived from the carotid sympathetic plexus and traverse the wall of the carotid canal to join the plexus.

The tympanic plexus supplies branches to the mucosa of the tympanic cavity, pharyngotympanic tube and mastoid air cells. It sends a branch to the greater petrosal nerve via an opening anterior to the fenestra vestibuli. The lesser petrosal nerve, which may be regarded as the continuation of the tympanic branch of the glossopharyngeal nerve, traverses the tympanic plexus. It occupies a small canal below that for the tensor tympani. It runs past, and receives a connecting branch from, the geniculate ganglion of the facial nerve. The lesser petrosal nerve emerges from the anterior surface of the temporal bone via a small opening lateral to the hiatus for the greater petrosal nerve and then traverses either the foramen ovale or the sphenopetrosal fissure or the innominate canal (of Arnold) to join the otic ganglion. Postganglionic secretomotor fibres leave this ganglion in the auriculotemporal nerve to supply the parotid gland. In former times, the operation of tympanic neurectomy was undertaken to reduce salivation in patients who drooled and to diminish facial sweating in those with Frey's syndrome following parotidectomy; less invasive and more effective procedures are employed nowadays.

Facial nerve

The facial nerve consists of intracranial (cisternal), intratemporal and extratemporal portions. The intratemporal portion is further divided into meatal, labyrinthine, tympanic (horizontal) and mastoid (vertical) segments (Fig. 37.14). The cisternal portion emerges from the ventrolateral aspect of the caudal border of the pons. The meatal portion enters the porus acusticus of the internal auditory meatus accompanied by the nervus intermedius, the vestibulocochlear nerve and the labyrinthine vessels (see Fig. 37.7). In this part of its course, the nerve lacks a fibrous sheath or endoneurium and is surrounded by a thin layer of arachnoid. The motor root, which supplies the muscles of the face, and the nervus intermedius, which contains sensory fibres concerned with the perception of taste and parasympathetic (secretomotor) fibres to various glands, are still separate components; they usually merge within the meatus. The labyrinthine segment runs from the fundus of the internal acoustic meatus to the geniculate ganglion, where the nerve makes the first bend or genu (see Figs 37.7, 37.13). The labyrinthine segment is the shortest and narrowest part of the facial nerve; it lacks anastomosing arterial cascades, which renders it susceptible to vascular compression. The tympanic part initially curves around the oval window niche, then lies just anterior and inferior to the lateral semicircular canal and bends again at the second genu to become the vertical or mastoid part. The mastoid part is the longest of the petrous segments and runs from the pyramidal process to the stylomastoid foramen. The tympanic and mastoid segments of the facial nerve are supplied by the facial arch, formed by the petrosal branch of the middle meningeal artery and the stylomastoid branch of the posterior auricular artery. The extratemporal course of the facial nerve is described on page 502.

The branches that arise from the facial nerve within the temporal bone can be divided into those that come from the geniculate ganglion and those that arise within the facial canal.

The main branch from the geniculate ganglion is the greater (superficial) petrosal nerve, a branch of the nervus intermedius. The greater petrosal nerve passes anteriorly, receives a branch from the tympanic plexus and traverses a hiatus on the anterior surface of the petrous part of the temporal bone. It enters the middle cranial fossa, runs forwards in a groove on the bone above the lesser petrosal nerve, and then passes beneath the trigeminal ganglion to reach the foramen lacerum. Here it is joined by the deep petrosal nerve from the internal carotid sympathetic plexus, to become the nerve of the pterygoid canal (Vidian nerve). The greater petrosal nerve contains parasympathetic fibres destined for the pterygopalatine ganglion, and taste fibres from the palate. The geniculate ganglion also communicates with the lesser petrosal nerve.

The nerve to stapedius arises from the facial nerve in the facial nerve canal behind the pyramidal eminence of the posterior wall of the tympanic cavity and passes forwards through a small canal to reach the muscle.

The chorda tympani (see Fig. 37.13; Fig. 37.15) leaves the facial nerve some 6 mm above the stylomastoid foramen and runs anterosuperiorly in a canal to enter the tympanic cavity via the posterior canaliculus. It then curves anteriorly in the substance of the tympanic membrane between its mucous and fibrous layers (see Fig. 37.15A), and

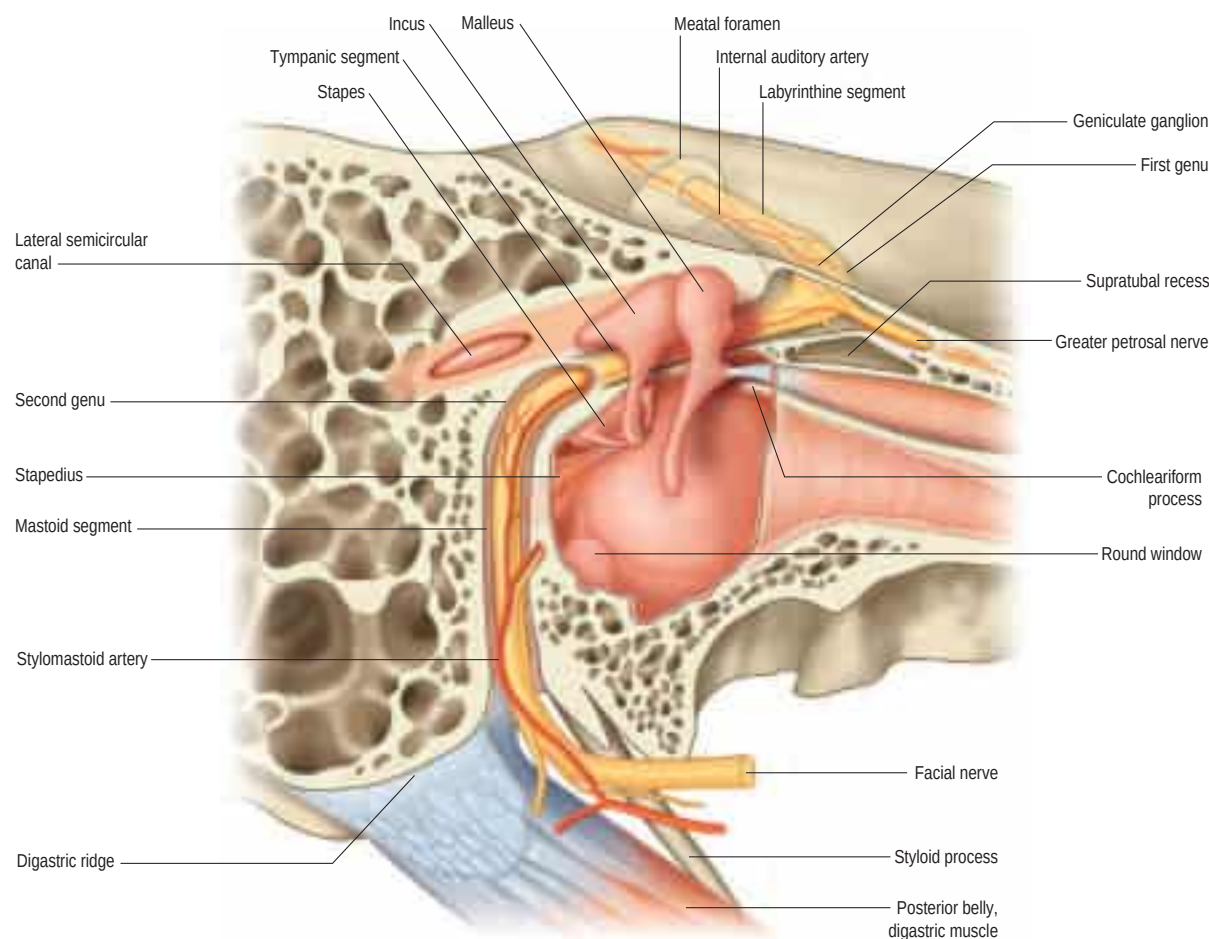


Fig. 37.14 The anatomy of the infratemporal portion of the facial nerve and associated middle ear structures. The nerve is vulnerable to injury at several sites. Perigeniculate region: nerve compression and ischaemia are most likely to occur in the narrow meatal and labyrinthine segments, which are the vascular watershed areas of branches of the external carotid artery and the posterior circulation; the first genu is tethered by the greater petrosal nerve, increasing susceptibility to shearing injuries; and the geniculate ganglion is susceptible to injury during surgical dissection in the supratubal recess of the anterior epitympanum. Tympanic segment: the nerve is most frequently dehiscent above the oval window; the distal tympanic segment and the second genu are both susceptible to injury in cholesteatoma surgery as a result of pathological dehiscence or distorted anatomy and failure to identify important surgical landmarks. Mastoid segment: the facial nerve is positioned lateral to the tympanic annulus and just distal to the stylomastoid foramen in the lower portion of its vertical course and is therefore susceptible to injury during surgery involving the external acoustic meatus. (Redrawn from Francis HW 2010 *Anatomy of the temporal bone, external ear, and middle ear*. In: Flint PW, Haughey BH, Lund VJ, et al (eds) *Cummings Otolaryngology: Head & Neck Surgery*, 5th ed. Mosby. Ch 127, 1821–1830.)

crosses medial to the upper part of the handle of the malleus to the anterior wall of the tympanic cavity, where it enters the anterior canaliculus. It exits the skull at the petrotympanic fissure, and its further course is described on [page 552](#). The chorda tympani contains parasympathetic fibres that supply the submandibular and sublingual salivary glands via the submandibular ganglion and taste fibres from the anterior two-thirds of the tongue.

Dehiscences of the facial nerve canal

The facial nerve may be somewhat variable in its anatomical course through the temporal bone ([Proctor and Nager 1982](#)). It may split into two or three strands, starting at the geniculate ganglion, and then make its way across the promontory to the stylomastoid foramen, or pass a few millimetres posteriorly to its second genu, before it turns inferiorly posterior to the fossa incudis, a position where it is particularly vulnerable during surgical exploration of the mastoid antrum. The more proximal the division into strands, the more bizarre is the subsequent course. More distal bifurcations pass either side of the fenestra vestibuli. It may be dehiscent, particularly in its second part, when it occasionally overhangs the stapes, or run inferior to the stapedial superstructure, a position that renders it vulnerable during surgery to the stapes ([Barnes et al 2001](#)). The motor fibres to the face may be carried through the chorda tympani, which is then enlarged. When this is the case, the distal facial nerve dwindles to a fibrous strand in a narrowed stylomastoid foramen. In chronic bone disease in the tympanic cavity, the facial nerve may be exposed in its canal. Inflammation may lead to facial paralysis of the infranuclear or lower motor neurone type.

Bell's palsy

Bell's palsy is the eponym given to an idiopathic lower motor neurone facial palsy. It may be complete or partial, and is characterized by a

flaccid paralysis of the ipsilateral muscles of facial expression; decreased lacrimation in the ipsilateral eye (which is controlled by neurones in the greater petrosal nerve); and hyperacusis or decreased tolerance of loud noises in the ipsilateral ear as a result of paralysis of stapedius. Its cause remains the subject of speculation. Magnetic resonance imaging studies suggest that there are inflammatory changes in labyrinthine and perigeniculate segments of the facial nerve. In the vast majority of cases, spontaneous full recovery takes place after a few weeks.

TYMPANIC MUCOSA

The mucosa of the tympanic cavity is pale, thin and slightly vascular. It is continuous with that of the pharynx, via the pharyngotympanic tube, and covers the ossicles, muscles and nerves in the cavity to form the inner layer of the tympanic membrane and the outer layer of the secondary tympanic membrane. It also spreads into the mastoid antrum and air cells. The middle ear mucosa is a mucus-secreting respiratory mucosa bearing cilia on its surface. The precise distribution of the mucociliary epithelium varies in normal middle ears, being more widespread in the young. Three distinct mucociliary pathways have been identified – epitympanic, promontorial and hypotympanic – the latter being the largest. Each of these pathways coalesces at the tympanic orifice of the pharyngotympanic tube ([Gleeson et al 1991](#)).

The mucosa forms several vascular folds that extend from the tympanic walls to the ossicles: one descends from the roof of the cavity to the head of the malleus and the upper margin of the body of the incus, and a second surrounds the stapedius. Other folds invest the chorda tympani nerve and tensor tympani. The folds separate off saccular recesses that give the interior of the tympanic cavity a somewhat honeycombed appearance; these mucosal folds have been described in greater detail by [Proctor \(1964\)](#). Of note, the superior recess of the

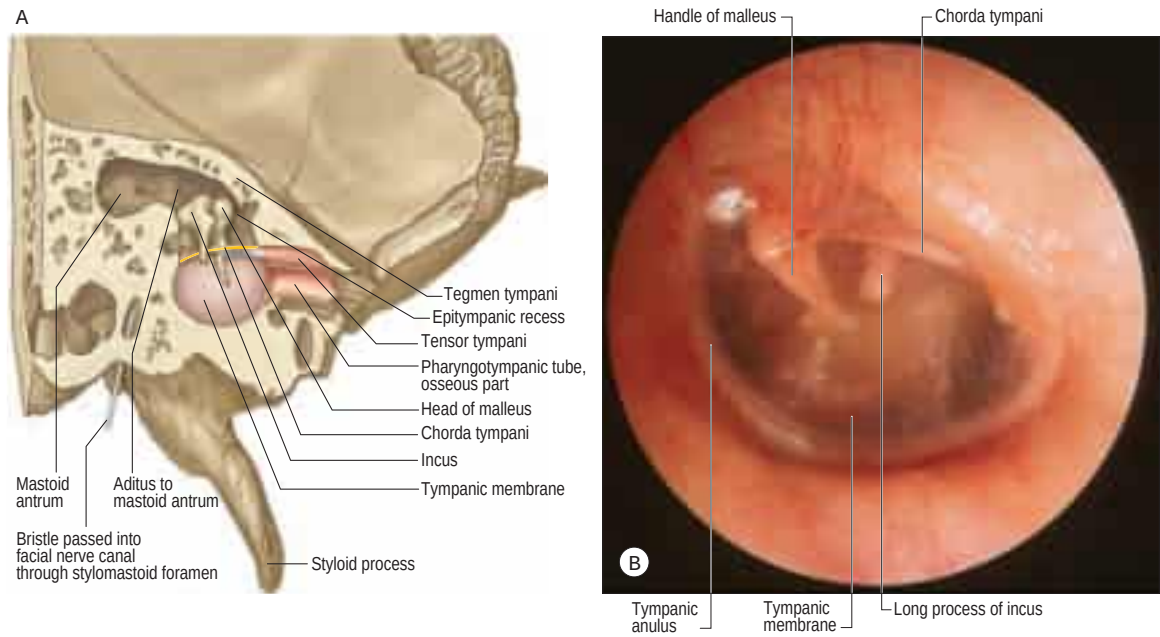


Fig. 37.15 The chorda tympani nerve. **A**, An oblique vertical section through the left temporal bone, to show the roof and lateral wall of the middle ear, the chorda tympani and the mastoid antrum. **B**, The chorda tympani crossing the tympanic membrane, auroscopic view. (B, Courtesy of Mr Simon A Hickey.)

tympanic membrane, Prussak’s space, lies between the neck of the malleus and the pars flaccida, bounded by the lateral malleolar fold. This space can play an important role in the retention of keratin and subsequent development of cholesteatoma.

Cholesteatoma

Cholesteatoma is the name given to keratinizing squamous epithelium within the middle ear. There is debate as to how such epithelium comes to be in the middle ear. Theories include development from embryological cell rests, metaplasia from inflamed mucoperiosteum, and aberrant migration of squamous epithelium either through a perforation in the tympanic membrane (usually in the pars flaccida or posterosuperior pars tensa) or within an area of tympanic membrane atelectasis where the tympanic membrane becomes adherent to the medial wall of the tympanic cavity. It is likely that all of these processes may be implicated at some time. A feature of cholesteatoma that remains poorly understood is its ability to erode bone, by activating osteoclasts, thus allowing the epithelium to proliferate and invade, destroying the temporal bone

and carrying infection to the soft tissues. Cholesteatoma can therefore cause deafness through damage to the ossicles and inner ear; problems with balance through damage to the vestibule and semicircular canals; facial palsy through ischaemia and necrosis of the facial nerve; and intracranial sepsis. Treatment involves microsurgical dissection of the invading sac of epithelium with, wherever possible, preservation of the delicate structures just listed.

Bonus e-book images and table

Fig. 37.9 C, D, E, Coronal CT scans showing details of the ossicular mass. **C**, Malleus and tensor tympani. **D**, Incus. **E**, Incus and incudostapedial joint.

Table 37.1 Common anomalies of the pinna.

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